

CYBERSECURITY THREAT DETECTION USING ANOMALY DETECTION AND MACHINE LEARNING TECHNIQUES

Subtitle if required

By

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Declaration

I hereby certify that this report constitutes my own work, that where the language of others is used, quotation marks so indicate, and that appropriate credit is given where I have used the language, ideas, expressions, or writings of others. I declare that this report describes the original work that has not been previously presented for the award of any other degree of any other institution.

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Abstract

Cyber security threat is continuously growing in terms of sophistication, thus making signature-based intrusions detection systems obsolete when dealing with zero-day threats, polymorphic threats and other advanced cyber security threats. The current research work is concerned with anomaly detection using machine learning techniques in order to enhance cybersecurity threat detection. The current research will conduct a review of the existing literatures on cyber security threat, anomaly detection, supervised and unsupervised machine learning techniques, among others. The literature provides insights into the pros and cons of conventional detection approaches and stresses the significance of intelligent learning models in today's intrusion detection systems. The design employs a quantitative experiment based on the CICIDS2017 dataset. Prior to building the models, data pre-processing, feature engineering, and feature selection are conducted. Both supervised algorithms – Logistic Regression, Decision Tree, Random Forest, and XGBoost, and unsupervised approaches – Isolation Forest, K-Means, and Autoencoder models are used for evaluating performance through measures like accuracy, precision, recall, F1-Score, ROC-AUC, and confusion matrix. These results show that ensemble learning approaches such as Random Forest and XGBoost have good classification capabilities to detect malicious traffic within a network, while unsupervised anomaly detection algorithms are efficient in identifying new attacks. The comparison proves the benefits of the combination of supervised and unsupervised learning approaches in improving detection abilities and decreasing false alarms. On the whole, it can be concluded that the use of machine learning in cybersecurity is efficient in improving intrusion detection and cybersecurity activities

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Chapter 1 Introduction

1.1 Problem Description, Context and Motivation

Problem Description

Traditional cybersecurity systems have limited capabilities for identifying newly emerging cyber threats because of their signature-based approach to cyber threat detection. These systems go wrong when they encounter zero-day attacks, polymorphic malware and complex forms of intrusion. Organizations continue to be susceptible to operational disruptions, data breaches, and losses of financial figures. This research aims to solve it, implementing Python based machine learning solutions such as Random Forest, XGBoost and Autoencoder models that can identify anomalies in network behavior, not signatures, and identify previous unknown attacks with signature-based methods.

Context

Cyberattacks are increasingly common and frequent throughout the world, across all industries. Larger networks, critical infrastructure, and cloud systems are under increased threat as attackers increasingly from nation-state backed actors, criminals and individual hackers attack organizations. With remote working and the increase in the use of the Internet of Things, the attack surface is a growing concern and security perimeter is no longer proving effective (Jain and Mitra, 2025). The CICIDS2017 contains realistic network traffic data with multiple categories of network attacks, such as DoS/DDoS (Denial of Service), PortScan, Brute Force, Infiltration and Web Attacks. This area of research aligns with and tackles core defence requirements in business, public sector, healthcare, and critical infrastructure areas.

Motivation

This research aims to leverage machine learning in threat detection for enhancing cybersecurity capabilities. Cybercrime is costing trillions of dollars worldwide and needs more and better methods of detection (Dahir *et al.* 2024). Human resources are reduced, and response time is enhanced, by using automated detection. The goal of the research is to build models with a high detection accuracy that produce a minimum number of false positives, for use in real-world security operations. A platform that suits the development and testing of these machine learning solutions is an ideal language for Python

1.2 Objectives

Aim

The aim of this project is to examine how well the techniques of anomaly detection and machine learning can enhance cybersecurity threat detection and assist with malicious activity identification in contemporary digital landscapes.

Objectives

- To implement and test ensemble machine learning methods such as Random Forest and XGBoost classification of network traffic as benign or malicious.
- To create and evaluate multi class classification models to detect specific attack categories DoS/DDoS, PortScan, Brute Force, Infiltration and Bot activities.
- To create and apply models for anomaly detection for detection of novel threats without labeled data, such as Isolation Forest, K-Means clustering, Autoencoder.
- To analyze and compare thoroughly the different models implemented using complete accuracy, F1 score, ROC-AUC and confusion matrix analysis.

Research questions

- RQ1: How accurately are anomaly detection techniques in identifying cybersecurity threats?
- RQ2: What are the best performing machine learning methods for cybersecurity threat detection?
- RQ3: What accurate can machine learning be in the detection of new or zero-day attacks?
- RQ4: What obstacles and boundaries do pose to machine learning-based anomaly detection systems?

1.3 Methodology

Research Design

The research type used is quantitative experimental design with the development and testing of machine learning models in cybersecurity threat detection. This research relies on CICIDS2017 dataset which is a benchmark of a full-featured network intrusion detection system network traffic dataset with both attack and normal traffic. This choice of the dataset mirrors that in the field of cybersecurity research, and confirms that the dataset is useful for machine learning applications.

Data Preprocessing and Feature Engineering

The preprocessing of the data is the first step of the experiment, where low-quality features in the data are eliminated and missing data is filled by interposition with the median (Sekhar Dey *et al.* 2024). Binary labels assign for supervised learning tasks. Use feature variance analysis to select the most useful features for modeling. Scaling techniques are used for numerical stability between algorithms.

Model Development

Three of the categories are implemented as models to test against. For binary or multiclass classification of supervised classifiers such as Logistic Regression, Decision Tree, Random Forest and XGBoost are applied. For anomaly identification without labelled data, following unsupervised detectors are used: Isolation Forest, K-Means clustering, and Autoencoder. The two models are systematically trained and tuned for the hyper parameters.

Testing and Evaluation Framework

Evaluation is conducted using the hold-out validation methodology, where 30% of the data is used as test data and 70% as training data. This is a way to avoid overfitting when evaluating. The measures of good performance consist of accuracy, precision, recall, F1 score, ROC-AUC and confusion matrix analysis (Khalaf *et al.* 2024). In multiclass tasks there are weighted/metrics per class that give insight into the strengths of the model. The false positive and false negative rates are given special attention for their operational implications.

Project Management and Technologies

The project has an iterative approach which follows Agile principles and three sprints are planned: data preparation, supervised model development, and unsupervised model development (Ajayi *et al.* 2024). Data manipulation is done with Pandas and NumPy in Python, whereas gradient boosting and traditional machine learning algorithms are handled by XGBoost and Scikit-learn, respectively, in Python. Deep learning is done using TensorFlow/Keras in Python.

1.4 Legal, Social, Ethical and Professional Considerations

This research relies on cybersecurity data that is publicly available, and it has a number of significant legal, social, ethical and professional implications.

- **Legal Considerations:** CICIDS2017 does not contain personally identifiable information, thus is not covered by data protection legislation. The data used in the research is publicly released data volitionally used for educational purposes, of which there is no legal violations. There is no real-life data that is collected for any organization, so there are no concerns about unauthorized access.
- **Social Considerations:** This research advance the ability in cyber security that foster social welfare. By improving threat detection, individuals, organizations, and community are protected from cybercrime. Results contribute to the creation of effective protections against financial fraud, identity theft and privacy issues, thus enhancing the overall cyber security environment.
- **Ethical Considerations:** The study conforms to the values of openness, accountability, and protection of intellectual property rights. All sources mentioned in this research have been cited. There are no human subjects involved in this research; hence, no need for ethical approval from the Ethics Committee. The results target the improvement of defense systems.

Professional Considerations: Researcher adheres to a standard, good practice methodology by documenting it thoroughly, allowing others to replicate and verify the study. Results present objectively with appropriate limitation discussions. The research communicates professional knowledge in the areas of cybersecurity and machine learning.

1.5 Background

Digital technologies have transformed and facilitated the ways in which organisations work, creating new opportunities with cyber security, and new risks. Today's highly sophisticated attacks, and zero day, in particular, are impossible to be defended with current defensive technologies like firewall and signature-based IDS. Machine learning is one of the most disruptive solutions to address any suspicious activity in network traffic as cybersecurity goes. These techniques are designed to learn how to recognize a baseline of the user's behavior, and will alert in the event that they detect an anomalous activity that may correspond to malicious behavior. This is helpful in the event that a brand-new threat is being discovered. The traditional classification methods are not capable to give better performance compared to ensemble learning methods like XGBoost and Random Forest

in the classification of benign and malicious network traffic. These algorithms avoid overfitting, and improve the performance for generalization in complex detection tasks. Unsupervised anomaly detection has complementary capabilities to detect unknown breaches without the need for labelled trainable data. Techniques such as Isolation Forest and Autoencoders can detect anomalies using any patterns significantly different from representation of normal behavior. There are still major challenges to face in cybersecurity's threat detection. The class imbalance phenomenon is that there are many more benign than malicious examples and training a model is more difficult and it's harder to obtain a good score on the minority attack categories. To solve these challenges, specialized techniques and evaluation methodologies should be used to ensure the effectiveness of practical deployment in real security operations.

1.6 Structure of Report

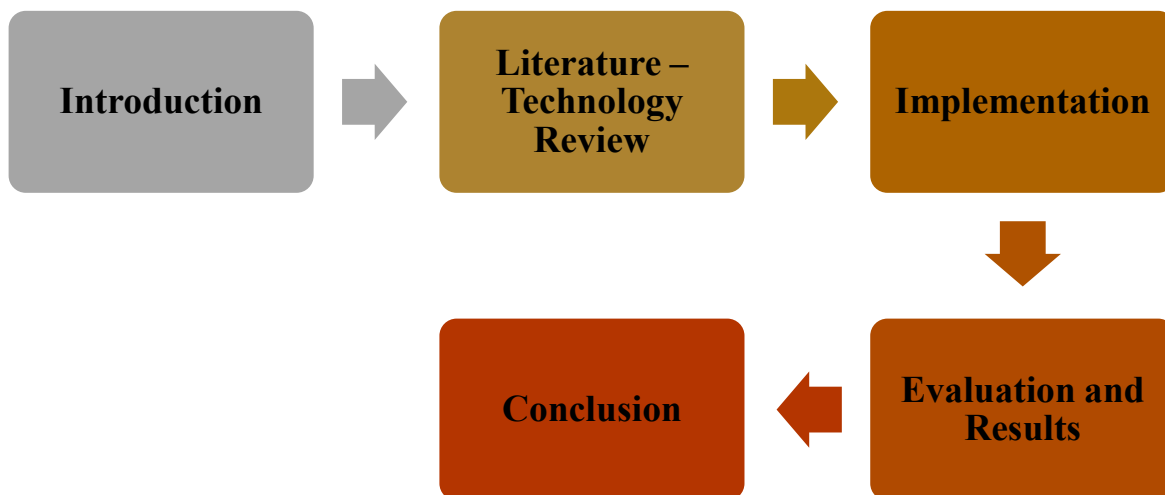


Figure 1.6.1: Structure of report

Chapter 2 Literature – Technology Review

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2.1 Literature Review

2.1.1 Evolution of Cybersecurity Threats and Traditional Detection Methods

The spread of rapid digital change, cloud-based systems, Internet of Things (IoT), and remote connections are adding to the complexity of organisational cybersecurity environments. This growing digital environment provides more access points for cybercriminals to attack, from ransomware, phishing and distributed denial of service (DDoS), to polymorphic malware and advanced persistent threats (APTs). Current cyber threat is invariably obfuscating and evading techniques which help the malicious code to evade the conventional security methods and undermining the static detection technique [6].

The conventional methods of protection are basically the use of firewalls, anti-virus and Signature based IDS systems. Firewalls control network traffic and perform checks on the security policy, and the AV provides for detection of malicious files by known malware characteristics. A zero-day attack is an attack at a time when no signature for it has been developed, so that signature-based detection has not work as there would be no signatures to detect it during the initial attack step [7].

Advanced cyber-attacks provide a challenge to rule-based security mechanisms because they can evade conventional defenses and affect previously unknown intrusions. The specificity of the attack signatures is overcome by anomaly-based detection, which looks for abnormal behavior rather than just relying on known attack signatures [8]. The approach of using intelligent detection techniques with the help of machine learning has received much attention as a viable solution in enhancing modern cybersecurity defence.

2.1.2 Anomaly Detection in Cybersecurity

Anomaly Detection is the action of detecting an activity that is outside of a normal pattern of system or network activity. This is an important method in cybersecurity that provides the ability to detect unusual activity that could be an attack, such as attempts to access without permission, malware running, inbound threats, and zero day breaches. Anomaly-based intrusion detection are more flexible to deal with the changing nature of attacks, since they are more oriented towards behavioural anomalies than towards known attack patterns [9].

Technique	Detection Principle	Strengths	Limitations	Cybersecurity Applications
Statistical Methods	Detect anomalies using probability distributions and predefined statistical thresholds.	Simple implementation, computationally efficient, suitable for stable network environments.	Performance declines when data distributions change or network behaviour becomes highly dynamic.	Network traffic monitoring, baseline anomaly detection, fraud detection.
Distance-based Methods	Measure the distance between observations and identify isolated instances as anomalies.	Effective for detecting outliers without requiring labelled data.	High computational cost when analysing large or high-dimensional datasets.	Intrusion detection, abnormal user behaviour analysis, network anomaly detection.
Density-based Methods	Identify anomalies by analysing the density of neighbouring data points within the feature space.	Detect complex and non-linear anomaly patterns with high accuracy.	Sensitive to parameter selection and computationally intensive for large datasets.	Zero-day attack detection, insider threat identification, advanced intrusion detection.

Table: 2.1.2.1: Comparison of Major Anomaly Detection Techniques in Cybersecurity

Anomaly detection methods are used for modern IDS. Probability distributions and thresholds are used with statistical methods to generate normal behavioural profiles and detect significant deviations by considering them as anomalies. There are distance-based approaches which compare similarities between data instances and act as outliers and

classify them as threats. These techniques offer complementary methods in analysing abnormal network activity in the various cybersecurity scenarios [10]. In dynamic environments, statistical methods are not as effective, but have good computational efficiency. Complex and non-linear patterns can be detected by density based methods but detection rate is influenced by parameter selection. Its integration of various anomaly detection methods enhances detection rates, minimizes false alarms, and increases the robustness of defense against sophisticated and novel cyber threats [11].

2.1.3 Machine Learning Techniques for Cybersecurity Threat Detection

Machine learning is also a critical part of today's cyber defense, powering the automated detection of malicious actions in the massive amounts of network traffic and system events. Machine learning improves intrusion detection by learning and identifying the complexity of network traffic, rather than relying on manually crafted rules and predefined signatures of intrusion [12]. The methods in the cyber field that use machine learning include supervised, unsupervised, semi-supervised learning and deep learning. For supervised learning, the objective is to train the models from labelled datasets, to classify normal activity and malicious activity with high prediction accuracy. Unsupervised learning looks at unlabeled data to reveal patterns and anomalies without learning about attack categories. Semi-supervised learning uses a small amount of labeled data and a large number of unlabeled data to learn effectively when there is little labeled cybersecurity data. The semi-supervised learning approach boosts the accuracy of the model when some of the data are labelled in the intrusion detection system context since it requires the use of both labelled and unlabelled cybersecurity data [13].

Technique	Data Requirement	Strength	Cybersecurity Application
Supervised Learning	Labelled data	High classification accuracy	Malware and intrusion detection
Unsupervised Learning	Unlabelled data	Detects unknown anomalies	Zero-day attack detection
Semi-supervised Learning	Limited labelled + unlabelled data	Effective with limited labelled data	Network intrusion detection

Deep Learning	Large datasets	Learns complex attack patterns	Advanced threat and malware detection
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Table 2.1.3.1: Machine Learning Techniques for Cybersecurity

Deep learning has the potential to go beyond machine learning by using multi-layer neural networks to “learn deeper” into highly dimensional cybersecurity data to uncover complex features. This ability also helps with the right detection of complex attacks, such as advanced persistent threats and zero-day exploits. Unlike conventional security methods, Intelligent learning models detect behavioural patterns and offer enhanced protection against evolving cyber threats, as they do not depend solely on attack signatures [14].

2.1.4 Supervised Machine Learning Algorithms for Intrusion Detection

The supervised machine learning algorithms have a crucial role in making decision as to whether the network traffic is normal or suspicious (malicious), as the algorithms learn from labelled network traffic. Rehman et al. (2025) found that supervised learning models outperform in detecting attacks by identifying the intricate relationships between the features of the network and accurately labelling and categorizing the attacks. It is well-understood due to its simplicity, quick training time and high interpretability that Logistic Regression is still widely used as a baseline model. Decision Tree algorithms offer interpretive decision rules, model non-linear relationships, but can be prone to overfitting with very deep trees. Random Forest which improves detection accuracy and reduces the variance of the model by aggregating multiple decision trees. Additionally, XGBoost was a highly accurate package that addresses the complexity of cybersecurity data in an efficient manner by way of gradient boosting. Another classification method with a good classification performance in high dimension data is the Support Vector Machine (SVM) classification, but SVM needs more computational resources for model training. Random Forest and XGBoost has been shown to outperform individual classifiers in detection performance on a variety of intrusion datasets [16].

Algorithm	Accuracy	Overfitting	Computational Cost	Key Advantage
Logistic Regression	Moderate	Low	Low	Simple and interpretable
Decision Tree	High	High	Low	Easy decision rules

Random Forest	High	Low	Medium	Robust and accurate
XGBoost	Very High	Low	High	Excellent predictive performance
Support Vector Machine	High	Low	High	Effective in high-dimensional data

Table 2.1.4.1: Comparison of Supervised Machine Learning Algorithms

The algorithms have different strengths and weaknesses. Logistic Regression is computationally efficient, but has poor performance in handling complex decision boundaries. Decision Trees provide interpretability, but are sensitive to noise in the information. Random Forest and XGBoost achieve greater accuracy but incur the cost of ensemble learning and come with higher computational demands due to the reduction in overfitting. It has found that none of the supervised algorithms consistently performs best in all cybersecurity setups, so model selection must be based on the characteristics of the datasets, detection goals, and computational resources [15].

2.1.5 Unsupervised Learning for Anomaly Detection

Unsupervised learning is an important technique for cyber security because it is not based on a labeled training set that defines normal network traffic. This ability helps identify unknown attacks and threats that fall outside existing classification techniques. The patterns in the network traffic are analysed by unsupervised algorithms, which allow detection of previously unknown malicious activities [18].

Autoencoders, K-Means, and Isolation Forest are common unsupervised methods for anomaly detection. The Isolation Forest method makes random partitions to isolate anomalous observations, and is efficient enough for use with large cybersecurity datasets. K-Means clusters similar network operations together, and labels the data points that fall far from the cluster centers as possible network anomalies. An autoencoder can compare the network traffic in a real-time network system with a reference and detect a complex attack pattern in high-dimensional network traffic with superior ability [20].

Technique	Strength	Limitation	Application
Isolation Forest	Fast anomaly detection	Less effective for subtle anomalies	Unknown attack detection

K-Means	Simple clustering	Sensitive to cluster selection	Network traffic clustering
Autoencoder	Detects complex anomalies	High computational cost	Zero-day and advanced threat detection

Table 2.1.5.1: Comparison of Unsupervised Learning Techniques

These techniques each have their own pros and cons. Isolation Forest is useful for detecting anomalies and can sometimes fail to detect subtle anomaly changes. K-Means is sensitive to the data distribution and is not as efficient when it has irregular distributions. Autoencoders have high detection accuracy take a more powerful computation resource and require more optimization of the model. Generally, it has been determined that the selection of an adequate unsupervised algorithm occurs based on considerations of the characteristics of the data, computational requirements, and also the complexity of the cyber threats [19].

2.1.6 Performance Evaluation of Machine Learning Models

The reliability and effectiveness of machine learning models used in cybersecurity threat detection are heavily dependent on performance evaluation. Some measures of accuracy are the ratio of correct examples classified, but just offer a rough assessment of the model's performance [22]. Precision assesses what percentage of attacks are accurately identified by the attacker in comparison to the total number of attacks detected; recall measures an attacker's ability to detect an attack successfully. F1-score is a ratio of the two scores, precision and recall that is used to balance the two when the class distribution is skewed. The ROC-AUC is a measure criterion of the discrimination ability of a classifier at various decision thresholds, while the confusion matrix provides detailed information about the number of true positives, true negatives, false positives and false negatives. In the context of cyber security, false positive and false negative rates are significant since false alarms can create potentially significant security alerts, or true attacks can go undetected [23]. Class imbalance of datasets can pose severe challenges for accuracy, with high accuracy potentially hiding low accuracy in minority attack classes. Using a set of evaluation metrics together yields a more accurate measure of the intrusion detection performances and permits a significant comparison of machine learning models [21].

2.1.7 Research Gap and Conceptual Foundation

The field of machine learning for cybersecurity threat detection has made significant strides recently with supervised learning, unsupervised learning, and deep learning. Previous research has been found to yield at least better detection accuracy and better identification of malicious network behaviour. Despite these progressions, there are still a number of drawbacks. Numerous studies have been conducted on various single machine learning algorithms with less comparative study of the supervised and unsupervised methods [26]. The problem of class imbalance persists, limiting the ability to catch minority attack types and the efficiency of cybersecurity operations by generating false positive alerts. Models have been identified as having limited generalisation in the different network environment, limiting their practical use [24]. The study of the various algorithms applied to intrusion detection in an extensive comparative study with more machine learning techniques may contribute in the selection of algorithm and may enhance the performance of intrusion detection. In the present work, these shortfalls are overcome and a holistic comparative assessment of various machine learning methods is made under standardized performance measurement conditions

2.2 Technology Review

Technology Options

Technological solutions have been implemented when it comes to the cybersecurity threat detection systems used. Python serves as the primary programming language in the field of machine learning, due to the available libraries for working with it. Another programming language is R, which also provides an option in the field of statistical computing. Java can also be used in enterprise-level systems. In the field of algorithms, Scikit-learn has many features for classical machine learning methods, such as Random Forest, Decision Tree, and Logistic Regression classifiers. There are also libraries in XGBoost that can be applied when implementing gradient boosting (Rao *et al.* 2024). The deep learning algorithms can be developed with TensorFlow and Keras for Autoencoder based solutions, and with PyTorch for another solution as well. In the fast-prototyping phase, Keras would be suitable and PyTorch would be suitable in the designing phase in terms of its flexibility. Manipulation of data uses Pandas for data structured and NumPy for calculations. Matplotlib and Seaborn can be used to use for visualization for understanding outcomes. Flask might be used in the deployment of API services.

Rationale for Technology Choice

The choice of Python is opted because of its vast field of libraries, its huge community of developers, and its ability to be cross-platform in nature. Scikit-learn allows the users to utilize the well-known algorithms irrespective of the API used. With respect to ensemble learning, XGBoost shows outstanding performance (Katiyar *et al.* 2024). In turn, TensorFlow and Keras allows AI/ML architectures to be developed very conveniently and effectively. All these technologies follow the directions of machine learning research and thus can be replicated and compared with other, more general, machine learning and cyber security research.

2.3 Summary

The literature survey have been given to make the reader familiar with the subject in general and to give understanding about how cybersecurity threat detection and how malicious network activities can be detected using Machine Learning techniques. The strengths, weaknesses, and relevance of traditional security mechanisms, anomaly detection approaches, supervised and unsupervised learning algorithms and performance evaluation metrics are critically reviewed. While previously published work has demonstrated good intrusion detection accuracy, issues of data class imbalancing, rate of false intrusions and the difficulties of model generalisation and comparison have shown up. In addition, the technology review has explored suitable technology to develop and process the data, including Python, Scikit-learn, TensorFlow, Keras, and XGBoost, all of which are suitable for cybersecurity research. This critical analysis has helped in finding out the gaps in the existing study and conceptual groundwork for the present study. The development of a research methodology, modeling, implementation and assessment of the model's performance in the following chapter

Chapter 3 Implementation

3.1 Introduction

The implementation of the proposed Cyber Security Threat Detection Framework is presented in this chapter based upon the methodology developed in Chapter 2 and with a structured experimental environment. The implementation of the conceptual research framework process into a practical analysis process involves systematic data preparation, model development and evaluation of model performance. For each stage of implementation, a pre-defined workflow is provided that facilitates analytical consistency, reproducibility and reliable research through each stage. This chapter focuses on pointing out how the adopted execution methods can support the development of an intelligent solution that can defend in cybersecurity in the controlled experimental environment. The chapter begins by an outline of the system design and implementation framework, then moves on to research approach, experimental environment and tools adopted in experimental analysis of the system in the perspective of implementation and execution. Then, the process of datasets preparation, data preprocessing, feature engineering and developing the machine learning models and in the final sections the training and validation process used in its implementation. Additionally, the chapter gives its highs and lows in implementation as well as what worked out and what didn't regarding the analytical development process. This is a structured implementation process that creates a logical sequence of the methodological foundation, which is presented in Chapter 2, and the comparative performance evaluation, which is discussed in Chapter 4.

3.2 Research approach

The implementation was adopted quantitatively to aid the systematic design and development of cyber threat detection framework. This together allowed for objective investigation on a numerical basis, as well as defined experiments and performance measurable. In quantitative research, patterns can be analyzed, anomalies can be investigated, and analytical processes can be evaluated and tested using statistics, but not intuition. The properties are very similar to those used in cyber security research, which relies heavily on the use of repeatable procedures, as well as empirical research to yield trustworthy and stable outcomes.

The research process adopted in data preparation, data refinement, framework construction, experimental implementation and results evaluation was organized and systematic. The given framework was evaluated using standard evaluation methodologies

which ensured objective evaluation and facilitated systematic comparison of the experimental results obtained. Recent research on cybersecurity acknowledges that quantitative research can be a successful method for the development and the validation of intelligent threat detection frameworks. It facilitates repeatable experiments under controlled analytical environment, objective analysis and evidence based decision making [29]. In addition, the quantitative approach that was used increased the reliability of the study due to minimizing personal bias and having valid findings based on evidence that would be measurable during the implement process.

3.3 System Design and Implementation Framework

The implementation framework created a framework of a well-defined workflow, which took raw network traffic and converted it into a quality analytic data set, appropriate to cybersecurity threat detection. A sequential design was used throughout each of the design implementation stages to maintain consistency for reliable data processing, for conducting control experiments, and for reproducing the experiments for evaluation [30]. The framework started from the CICIDS2017 dataset, as it is an updated one which reflects modern network traffic in a realistic enterprise environment and multiple ranges of cyberattacks. This contributed to make the implementation more relevant for the real intrusion detection and lead to a systematic analysis of the data. With data acquisition completed, the next step in the implementation is to process the data collected and perform the proper pre-processing steps. The importance of data quality has been brought to the fore by identifying missing data, non-unique data items, variables with inconsistent information and empty variables. This data transformation standardized the variables across all data to the same numeric format which allowed more analytical processing of the data in an efficient manner [31]. Feature engineering identified relevant traffic features to describe traffic communication behavior, traffic statistical properties, and network flow patterns, and feature selection removed redundancy and selected the attributes most useful for identifying threats.

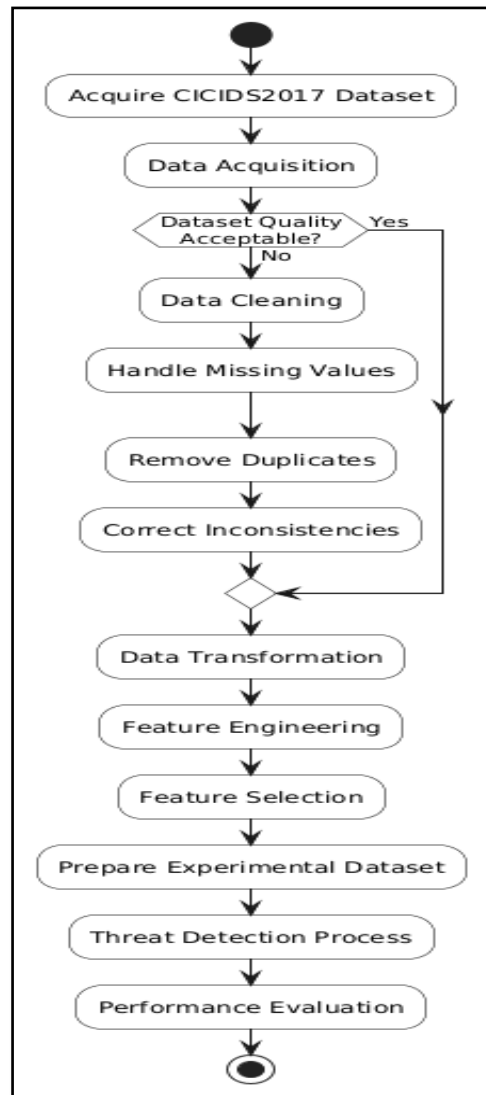


Figure 3.3.1: Research Workflow for the Implementation Process

The overall architecture is a linear data-flow type of the architecture that is adopted in the system where every stage of implementation provides validated data to the coming stage. This structure minimized the complexity of the experiments, ensured that data was kept in a consistent manner, and allowed for easy execution of a consistent workflow for the experiments. In this work, the research workflow involved in both the dataset preparation and feature refinement tools has been integrated into the same environment followed by implementation, prediction generation, performance evaluation and comparative analysis [32]. Such an implementation process brought about more repeatability, simpler checking and uniformity of the entire experimental process. Additionally, a transparent link between each step in the experiment processing level and the end-assessment process was also introduced to ensure standardized experimental results and cybersecurity good practices.

3.4 Experimental Environment and Development Tools

The implementation environment consisted of a general software collection and a super computing system that was used to process the data efficiently, develop experiments and analysis for these experiments during their research. The programming language was mainly Python as it enabled vast amount of support to writing data science, scientific computing and machine learning applications. As a consequence of the flexible design of the proposed solution, all the activities listed above, from data collection to the performance evaluation of the implemented solution, were carried out during one analytical session. As already mentioned above, that helped to improve the efficiency of the coding system and the process of organizing code execution [34].

The working environment was interactive (Jupyter Notebook), and the process of implementation was carried out in several analytical stages. Execution, documentation, debugging and verification of code were easy with notebook interface, whereby the code can be monitored throughout the implementation. Whereas with pandas, they were able to manipulate and transform the data without any hassle, and NumPy facilitated efficient high-level numerical computations with large-scale data, important aspects of analytical work. This integrated package, scikit-learn, was equipped with some functions for preprocessing, selecting features and implementing machine learning algorithms. Optimised computational framework TensorFlow and Keras were used to build the Deep Learning architectures, and Matlab and Seaborn were used for visualisation of data properties and analytical results. By integrating these libraries together, a standard implementation workflow was created, which further improved the computational performance, analytical understandability and experimental repeatability of cybersecurity research [33]. Implementation run on a computing platform using an Intel Core processor, 16 GB of memory and the Windows 11 operating system. This set up offered adequate computing power for the iterative data processing, efficient use of the memory space, and stable execution of the entire analytical process..

3.5 Dataset Collection and Understanding

The implementation was based on the CICIDS2017 dataset which was the main source of network traffic information for developing and testing the proposed cybersecurity threat detection framework. In addition to capturing all legitimate activities, this dataset contains a variety of types of malicious activity to simulate an enterprise network. Its detailed and fully defined set-up allows to systematically explore suspicious network activities in well-

controlled experiments with detailed network flow data and several attack scenarios. Using this dataset reinforced the relevance of the implementation due to the inclusion of representative data to represent the modern network and a changing cybersecurity landscape [35].

First, an examination has been conducted to possibly understand the overall nature of the data without analytical processing. This step included examining all of the available attributes, data structure, traffic categories, feature composition, and class distribution to get a good feel for the information available. A special focus was put on detecting the heterogeneity of network behaviours, and in acknowledging differences between the normal and malicious network behaviour. This initial assessment verified the complexity of the data, variety of the features and analytical needs for the following phases in implementations [36]. Additionally, the extensive data allowed for uniform implementation during testing and provided a solid foundation for further development of features, data preparation and analysis of experiments.

3.6 Data Preprocessing and Feature Engineering

The processing of data and the development of the features is an important aspect of the implementation process and it is quite essential for getting a structured representation of the raw network traffic data. A systematic approach to the improvement of the quality of data helps in overcoming inconsistency issues and enhancing the reliability of the subsequent experiments. For each pre-processing, the activities were shown in a particular order, which helps in boosting the uniformity in analyzing the cybersecurity threat detection framework, the efficiency of computing, and the completeness of the activities in each order [38].

Data Cleaning and Data Preparation

The implementation started with verifying and scrubbing the captured data in-depth to improve the quality of the captured network traffic data. The missing observations were analysed with great detail in order to reduce the impact of missing data on the processing of the analysis. Duplicate records are also removed to eliminate the need to compute duplicate observations because of the unnecessary computing overhead. These categorical attributes then were numerically coded, to provide a uniform representation across the model which was compatible with computational analysis. The numerical features were then normalised to reduce inter-range variations to give equal contributions

in the implementation of the features. Such pre-processing steps ensured consistency of the data sets, reduced noise and offered a foundation for the further processing of the data.

Feature Engineering and Feature Selection

The process of feature engineering attempted to make certain simplifications of the network traffic information in order to develop features that would be able to better analyze communication behavior and traffic. This made it possible to enhance the descriptive capabilities of the data by eliminating any nuisance variables from the data. The choice of features also enabled to identify informative features and eliminate redundant and highly correlated features that do not carry much analytical value. The process of reducing the feature space made it possible to improve computational efficiency and the quality of the experimental data set [37].

Dataset Partitioning for Experimental Analysis

The partition of the cleaned data set was performed in the last stage of pre-processing as two independent sets of training set and test set. This enabled way in which it would be possible to develop analysis, while at the same time being able to assess its performance independently. The use of this type of partition allowed a minimal bias in evaluation and increased the validity of the experiment. The entire preprocessing phase formed a solid basis for effective analysis and cybersecurity threat detection at all experimental levels.

3.7 Machine Learning Model Development

The creation of machine learning model was deemed to be the key stage of the implementation process because it entailed conversion of the created dataset into a smart system used for detecting cyber-attacks. There was a certain methodology of creating such a model which provided consistent application of all analysis tools with regard to the same parameters, training process, prediction creation and evaluation [41].

Supervised Learning Model Implementation

The implementation adopted various supervised learning approaches through Scikit-learn and XGBoost libraries within the Python programming environment. Each analytics approach was independently initialized after processing of the data set preparation and feature selection stage. The hyperparameters of each analytics approach were set following similar experimental conditions in order to ensure fair analysis throughout the

whole implementation process. The training process involved the use of the training data set such that each analytics approach could be able to detect patterns corresponding to legitimate and malicious traffic in the network. The prediction phase involved the testing data set in order to provide classification output for evaluation purposes[39].

Unsupervised and Deep Learning Model Implementation

The implementation also used unsupervised learning and deep learning approaches for determining unusual activity on the network based on a different approach to analysis. The configuration involved setting parameters, identifying anomalies and making predictions based on the preprocessed data. In the deep learning model, the configuration involved using an encoder-bottleneck-decoder method that identified representative patterns in network traffic based on features compression and reconstruction. The reconstruction error helped in determining unusual activities by identifying instances of deviations from normal behavior. This helped in achieving reproducibility in analysis and also comparison of the whole cybersecurity threat detection approach [40].

Implementation Workflow

The full implementation was built on a step-by-step procedure which involved the model initialisation, setting up parameters, training, prediction making and performance evaluation.

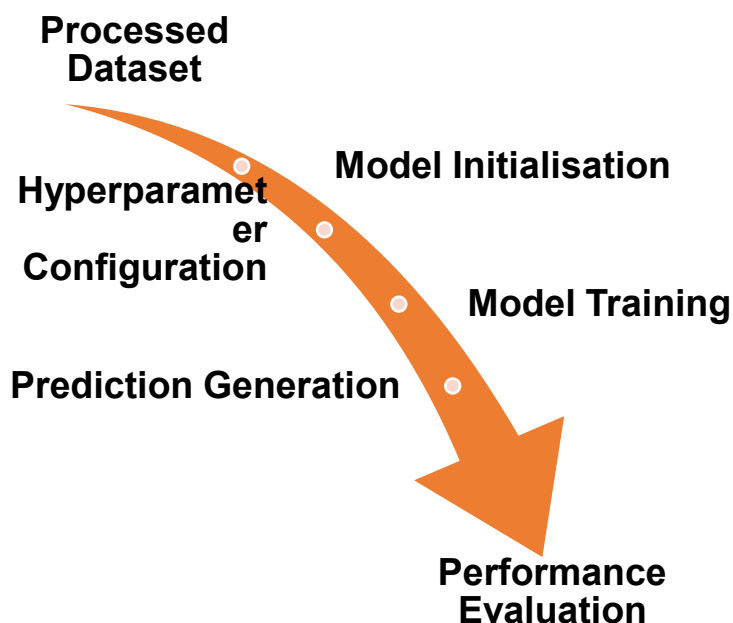


Figure 3.7.1: Machine Learning Model Development Workflow

The systematic implementation process used for creating the machine learning (ML) model. The process goes through a sequential procedure of model initialisation, parameter setup, training, prediction generation, and performance evaluation; which guarantees the analytical coherence, reproducibility and reliability of the procedure, throughout the study conducted. A standard procedure was followed to ensure the integrity of analysis, reduce implementation bias and enable consistent analysis across chapters in Chapter 4.

3.8 Model Training, Validation and Performance Evaluation

The last step of implementation was the model training, validation and testing of the model to check on the ability of the successful cybersecurity threat detection framework to perform analysis of the threats. The structured evaluation workflow enabled the independent testing as well as the use of standard assessment tools to ensure objective assessment, analytical reliability and reproducibility of the process for implementation [43].

Model Training and Testing Process

After preprocessing and feature engineering, the implementation began with the training of the models with the prepared training dataset. The entire experimental process was carried out under the premise of being fair and freezing the experimental data so that each analytical technique had the same input and parameters. Training allowed detection of network traffic patterns to be seen for both legitimate and malicious activities. After learning, the independent test set produced prediction results without having a negative impact on the creation of the model. This separation of training from testing meant that there was no evaluation bias and analysis was being properly tested under similar experimental conditions [42].

Validation Process

The validation stage was concerned with the quality of the predictions, which were checked through conducting the predictive classification and comparing it with the actual traffic tags while the network was being used. Validation of the work independently enhanced the reliability of the analysis through checking whether every used approach would be capable of generalizing the training data. The verification process enhanced the uniformity of the experiment, the reliability of the analysis and comparison analysis.

Performance Evaluation Metrics

The assessment involved a set of assessment tools to measure performance and provide a holistic analysis skill overview. For accuracy, the overall classification correctness was used, for precision, the percentage of successful attack predictions was used. Recall was a measure used to detect real malicious activities, and F1-score was employed as a compromise measure to show the overall performance of both the precision and recall. ROC-AUC represented the ability of the discrimination at different classification levels while the confusion matrix summarised the result of the prediction in terms of true positive, true negative, false positive and false negative classifications.

3.9 Implementation Challenges and Solutions

Implementing procedure had several technical issues that resulted in the compromising of data quality, computational time and reliability of analysis. All problems are addressed systematically to assure the same way of execution, credibility of the experiment and detection of cyber threats.

Challenge 1: Class Imbalance

Network traffic distribution was imbalanced with a mixture of normal and malicious observations, leading to the potential for skewed analytical learning from the network traffic if the observation distribution is skewed towards the major classes. This problem is addressed by carefully pre-processing the data and identifying the features that make the data informative and facilitate analytical development [44].

Challenge 2: High-Dimensional Network Data

The dataset contained many traffic characteristics that made calculations more difficult and complicated, and consequently, helped to mask the analysis. The available data was enhanced by feature engineering, which led to the generation of certain sets of representative traffic characteristics, as well as by feature selection to identify non-redundant features and find the most effective features for cybersecurity analysis. This helped in better analytical execution, efficiency of computing and not hampered information quality [45].

Challenge 3: Computational Training Time

A significant amount of computing resources was generally consumed during the analytical training in order to perform a large scale network traffic. A framework has been developed to implement them and how to set and remove any unwanted features, which reduces the

processing load and improves the processing efficiency. It was also involved in the analytical stability of the experiments' development in a strict procedure in all the experimental processes.

Challenge 4: False Positive Detection

A lack of traffic classification and identification has created the risk of a false security alarm. This condition was circumvented by the use of comparative evaluations of the same specimens with several methods of analysis. The same strategies demonstrated to remove unnecessary false alarms could be adapted to those that removed the need to detect a threat..

Challenge 5: Experimental Consistency and Reproducibility

For correct comparison, it is important to maintain similar implementation parameters for all of the analytical methods. The experimental evaluation was reasonable, parameter configuration for each experiment was the same and uniform preprocessing and partitioning of datasets. The aim of this structured implementation strategy was to increase the level of reproducibility, decrease implementation bias, and a solid base for comparative performance evaluation presented in Chapter 4.

3.10 Chapter Summary

The implementation of the proposed framework for detecting cyber security threats was covered in the chapter; thus, the methodology became the base for developing the framework. The implementation of the framework included the following steps – system design, research approach, experimental environment and development tools that made the research possible. The next step was to collect the data, preprocess it, engineer features and select the appropriate ones for getting the high-quality input data for further analysis. The implementation of a good program for training, validation, and performance evaluation helped in providing a great framework in which the ability of detection can be analyzed on the basis of different performance measures like accuracy, precision, recall, F1 Score, ROC-AUC, and Confusion Matrix. Moreover, the chapter discussed the possibility of adoption in order to solve the implementation issues and ways to make the analysis more reliable. The process of implementation of these phases has provided a good basis for experimentation, which ensure an objective performance evaluation. The findings from the experiments done using the models used in this chapter and the results of the performance evaluation are provided in the following chapter

Chapter 4 Evaluation and Results

4.1 Introduction

This chapter provides a complete suite of experiments to test and evaluate the algorithms for detecting cybersecurity threats based on anomaly detection and machine learning. The analysis starts by describing the data, giving details of data characteristics, class distributions, and any data preprocessing done. The analysis of the results involves three different investigations: supervised binary classification, supervised multiclass classification and unsupervised anomaly detection. Multiple performance indicators such as accuracy, F1 Score, ROC – AUC, confusion matrix analysis are used for judging the evaluation. The chapter ends with a discussion of the important findings, implications for Cyber Security applications and summary of experimental results.

4.2 Dataset Description

The data used in this research comes from a network traffic data set extracted from the collection CICIDS2017, which includes a set of eight different trace files collected during several days. The 2,830,743 instances in this dataset include 80 features, creating a complete cybersecurity benchmark. There is a very uneven class distribution with benign traffic representing 80.3% and attack traffic representing 19.7% of the observations. According to the malicious categories, DoS/DDoS attacks are the most on the record with 380,688, while PortScan is second with 158,930. There are minority classes that come up with very few instances: Infiltration only has 36 instances and Web Attack only has 11 observations. There are 54 integer variables, 24 float variables and 2 object variables. The Flow Bytes/s metric shows very few missing data points with only 1,358, which is ideal for machine learning applications. A realistic traffic set with a number of realistic attack patterns over various time periods for researchers to test and validate supervised and unsupervised detection methods.

4.3 Result Analysis

```
— Label distribution (raw) —
Label
BENIGN                2273097
DoS Hulk              231073
PortScan              158930
DDoS                  128027
DoS GoldenEye         10293
FTP-Patator           7938
SSH-Patator           5897
DoS slowloris         5796
DoS Slowhttptest      5499
Bot                   1966
Web Attack ♦ Brute Force 1507
Web Attack ♦ XSS        652
Infiltration          36
Web Attack ♦ Sql Injection 21
Heartbleed            11
Name: count, dtype: int64

— Data types overview —
int64    54
float64   24
object    2
Name: count, dtype: int64

— Missing values per column (top 20) —
Flow Bytes/s          1358
Destination Port       0
Total Fwd Packets      0
Flow Duration          0
Total Length of Fwd Packets 0
Total Length of Bwd Packets 0
Fwd Packet Length Max  0
Total Backward Packets 0
Fwd Packet Length Min  0
Fwd Packet Length Mean 0
Bwd Packet Length Max  0
Fwd Packet Length Std  0
Bwd Packet Length Min  0
Bwd Packet Length Mean 0
Bwd Packet Length Std  0
Flow Packets/s         0
Flow IAT Mean          0
Flow IAT Std           0
Flow IAT Max           0
Flow IAT Min           0
dtype: int64
```

Figure 4.3.1: Label Distribution, Data Type Overview, Missing Values

The data set comprises 2,830,743 data items, 2,273,097 of which are benign, and 557,646 of which are attacks. Contains 54 integers, 24 floats, and 2 object variables. The Flow Bytes/s feature only had 1,358 missing values, making it a good feature for machine learning models that will need minimal preprocessing.

```

— Attack category distribution —
Attack_Category
BENIGN          2273097
DoS_DDoS        380688
PortScan        158930
Brute_Force     13835
Other            2180
Bot              1966
Infiltration     36
Web_Attack       11
Name: count, dtype: int64

```

Figure 4.3.2: Cybersecurity Attack Category Distribution

Benign traffic dominates, with 2273097 observations (80.3%), followed by DoS/DDoS attacks with 380688 observations, PortScan with 158930 observations and Brute Force with 13835 observations (Salman *et al.* 2025). In particular, smaller classes like Infiltration (36) and Web Attack (11) show significant class imbalance that needs to be carefully considered during model evaluation.

```

# Strip whitespace from column names
data.columns = data.columns.str.strip()

# Drop non-feature columns
drop_cols = ['Label', 'Attack_Category', 'Binary_Label', '_source_file']
feature_cols = [c for c in data.columns if c not in drop_cols]

# Convert all feature columns to numeric
for col in feature_cols:
    data[col] = pd.to_numeric(data[col], errors='coerce')

# Drop columns with >50% missing or constant columns
missing_ratio = data[feature_cols].isnull().mean()
constant_cols = [c for c in feature_cols if data[c].nunique() <= 1]
drop_these = list(missing_ratio[missing_ratio > 0.5].index) + constant_cols
feature_cols = [c for c in feature_cols if c not in drop_these]
print(f"\nDropped {len(drop_these)} low-quality columns. Remaining features: {len(feature_cols)}")

# Fill remaining NaNs with median and replace inf values
data[feature_cols] = data[feature_cols].replace([np.inf, -np.inf], np.nan)
data[feature_cols] = data[feature_cols].fillna(data[feature_cols].median())

# Save processed references for next sections
X_raw = data[feature_cols].copy()
y_binary = data['Binary_Label'].copy()
y_multi = data['Attack_Category'].copy()
label_raw = data['Label'].copy() if 'Label' in data.columns else data['_Label'].copy()

print(f"\n✅ Preprocessing complete. X shape: {X_raw.shape}")
print(f"    Binary class balance:\n{y_binary.value_counts()}")

Dropped 8 low-quality columns. Remaining features: 70

✅ Preprocessing complete. X shape: (2830743, 70)
Binary class balance:
Binary_Label
0    2273097
1     557646
Name: count, dtype: int64

```

Figure 4.3.3: Preprocessing of Cybersecurity Data for Anomaly Detection

A total of 8 low-quality features have been eliminated during pre-processing, with 70 out of 71 quality features remaining as features for prediction, and with 2830743 observations remaining. The data has been cleaned for supervised and anomaly detection models, with missing values filled in with the median value and binary labels assigned, to ensure the resulting data is usable. To ensure that the resulting data has been used, missing values have been filled in with the median value, and binary labels have been assigned.

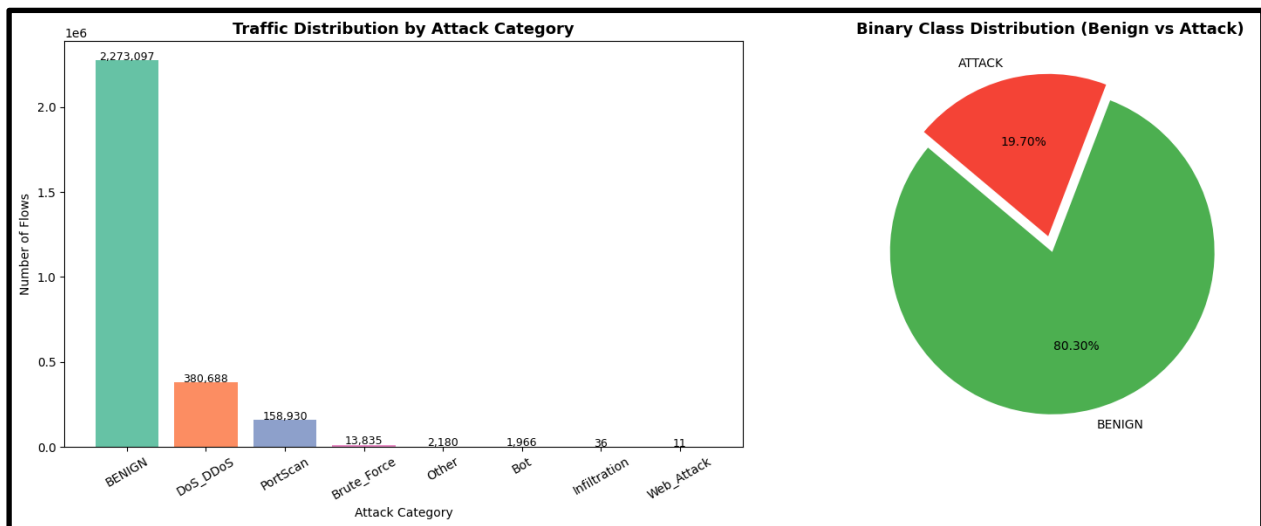


Figure 4.3.4: Traffic Distribution by Attack Category and Binary Class Distribution

The dataset is mostly dominated by benign traffic (80.30%), while the attack traffic in the dataset is 19.70%. Of the attacks, DoS/DDoS (380,688) is the largest category, significantly outnumbering PortScan (158,930) and Brute Force (13,835), which represents a typical imbalance between ports and the network in real-world cybersecurity scenarios.

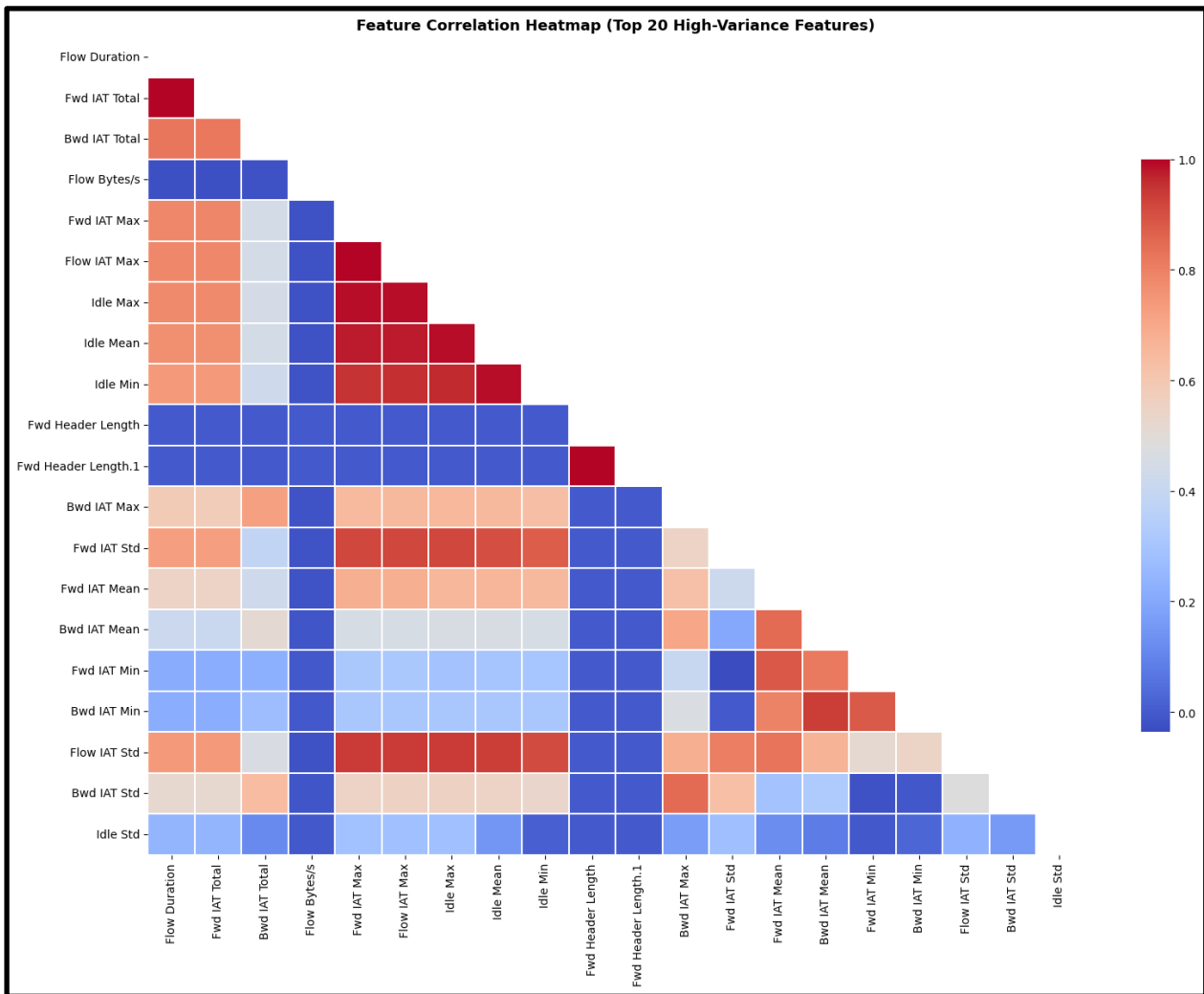


Figure 4.3.5: Feature Correlation Heatmap of Cybersecurity-Related Features

The packet length variable shows strong positive correlations in this correlation heatmap with the flow timing variables, with some coefficients being close to 1.0 and the other variable set around -0.6 . These connections suggest that the features are redundant, and that further feature selection is justified in order to have computational efficiency and good generalisation performance of the model.

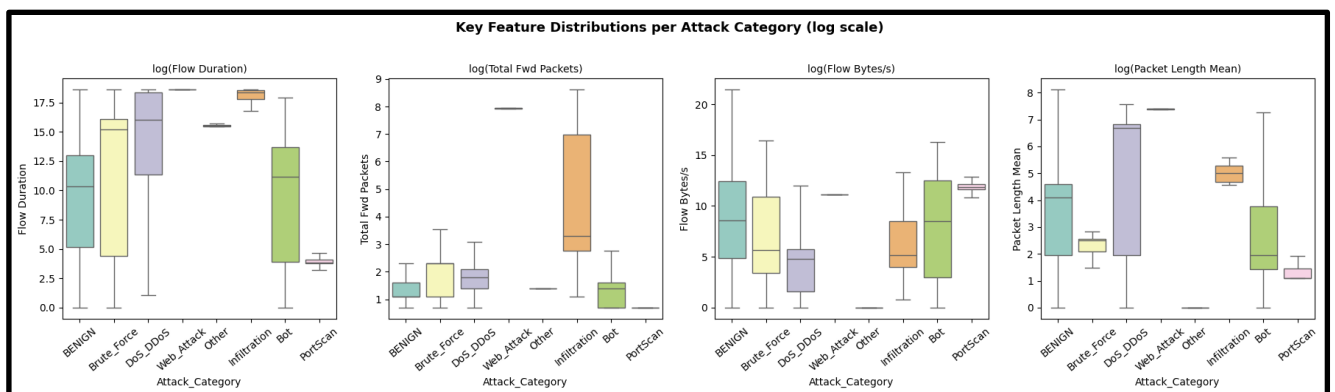


Figure 4.3.6: Anomaly Detection in Cybersecurity Features

Flow Duration, Flow Bytes/s, Flow Packets/s, and Packet Length Mean show differences in distribution among the attack categories through boxplots (Mizanur *et al.* 2025). It can be seen that Attack Categories have more variations and higher extreme values than Benign traffic, showing these variables have high discriminative ability for anomaly detection and classification.

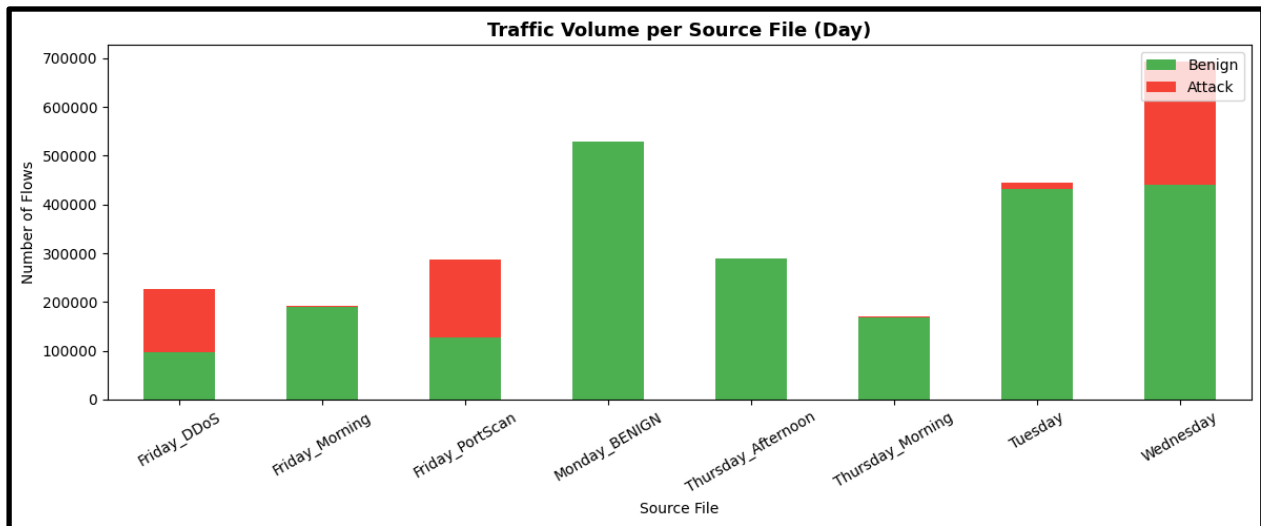


Figure 4.3.7: Traffic Volume per Source File

The nature of traffic varies significantly on collection days. The majority of the data from Monday has substantially benign traffic, high numbers of over 500,000 flows, while Friday afternoon has the highest concentration of attacks, specifically DDoS and PortScan data, showing that a wide range of network behaviours are successfully included to develop the models

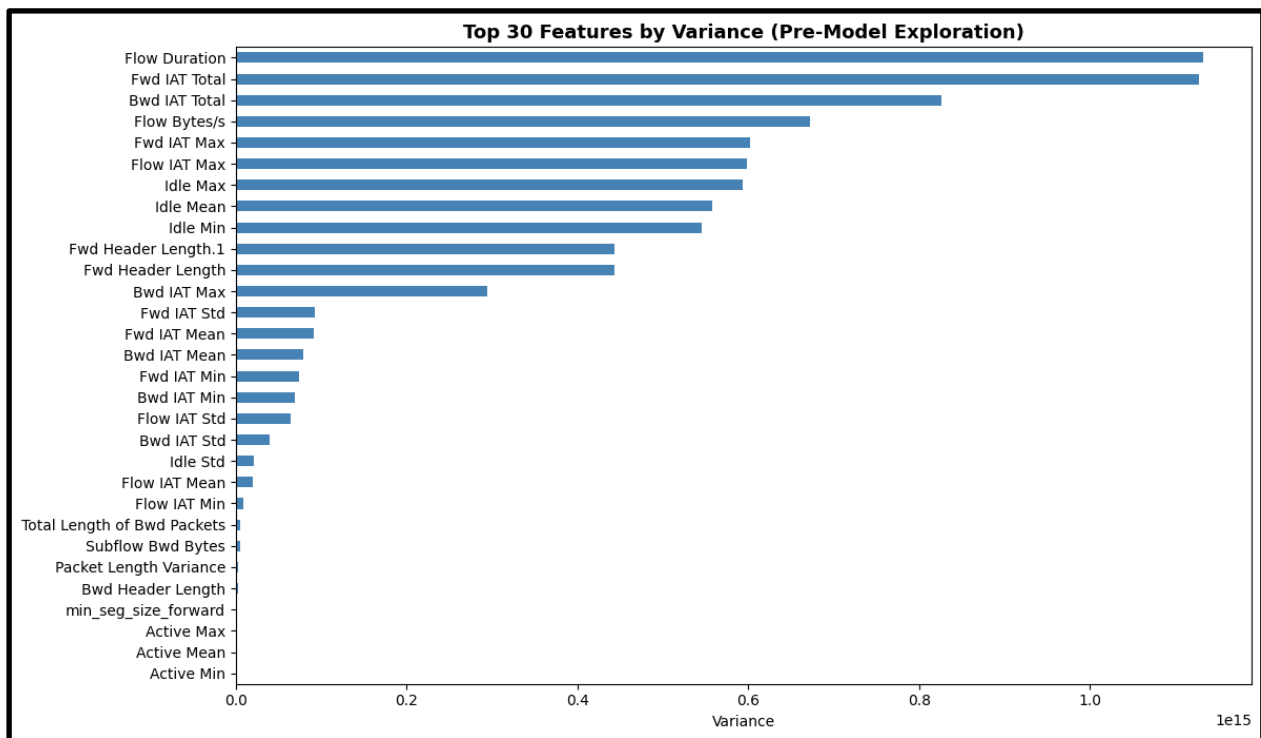


Figure 4.3.8: Feature Importance by Variance

Feature variance analysis revealed that Flow Duration, Flow IAT Total, Flow IAT Std, and Flow Bytes/s are the most informative features since their variances are close to 1.1×10^{15} (Okoli *et al.* 2024). But several characteristics of the communication service with a comparatively low variance and prediction value are not active, but they are rather passive characteristics of the message traffic.

— Logistic Regression —				
Accuracy:	0.9153			
F1 (weighted):	0.9198			
ROC-AUC:	0.9785			
Train time:	73.03s			
	precision	recall	f1-score	support
BENIGN	0.99	0.90	0.94	113655
ATTACK	0.71	0.96	0.82	27882
accuracy			0.92	141537
macro avg	0.85	0.93	0.88	141537
weighted avg	0.94	0.92	0.92	141537
— Decision Tree —				
Accuracy:	0.9944			
F1 (weighted):	0.9945			
ROC-AUC:	0.9984			
Train time:	27.39s			
	precision	recall	f1-score	support
BENIGN	1.00	0.99	1.00	113655
ATTACK	0.97	1.00	0.99	27882
accuracy			0.99	141537
macro avg	0.99	1.00	0.99	141537
weighted avg	0.99	0.99	0.99	141537

Figure 4.3.9: Logistic Regression and Decision Tree Binary Classification Report for Attack

Logistic Regression achieved 91.53% accuracy, 0.9198 F1-score, and 0.9785 ROC-AUC. Performance has been significantly boosted by Decision Tree, with an accuracy of 99.44%, an F1-score of 0.9945 and an ROC-AUC of 0.9984, meaning it has been able to neatly differentiate between benign and malicious network traffic while making only minor wrong classifications.

```

— Random Forest —
Accuracy:      0.9983
F1 (weighted): 0.9983
ROC-AUC:      0.9999
Train time:   266.59s
precision      recall  f1-score  support
  BENIGN        1.00    1.00    1.00    113655
  ATTACK        0.99    1.00    1.00    27882

accuracy
macro avg      1.00    1.00    1.00    141537
weighted avg   1.00    1.00    1.00    141537

— XGBoost —
Accuracy:      0.9986
F1 (weighted): 0.9986
ROC-AUC:      0.9999
Train time:   21.32s
precision      recall  f1-score  support
  BENIGN        1.00    1.00    1.00    113655
  ATTACK        0.99    1.00    1.00    27882

accuracy
macro avg      1.00    1.00    1.00    141537
weighted avg   1.00    1.00    1.00    141537

```

Figure 4.3.10: Random Forest and XGBoost Binary Classification Report for Attack

Random Forest lets the viewer know that accuracy is 99.83%, F1 score is 0.9983, and ROC-AUC is 0.9999, while XGBoost performs marginally better with 99.86% accuracy, 0.9986 F1 score, and 0.9999 ROC-AUC (Inuwa and Das, 2024). The two ensembles exhibited very high predictive accuracy and almost 100% cyberthreat classification.

SUMMARY TABLE					
	Accuracy	F1 (weighted)	ROC-AUC	Avg Precision	Train Time (s)
Logistic Regression	0.9153	0.9198	0.9785	0.9154	73.03
Decision Tree	0.9944	0.9945	0.9984	0.9888	27.39
Random Forest	0.9983	0.9983	0.9999	0.9996	266.59
XGBoost	0.9986	0.9986	0.9999	0.9997	21.32

Figure 4.3.11: Summary of All Model Results in Binary Classification

Comparing the performance, Logistic Regression (91.53%) methodology showed a lack of improvement, while the methodologies of Decision Tree, Random Forest and XGBoost classifiers showed a better performance (99.44%, 99.83%, 99.86% respectively). All models of Ensemble learning had the highest F1 score (0.9983 to 0.9986), and the highest ROC-AUC value (0.9999), indicating it is the most reliable model for classification.

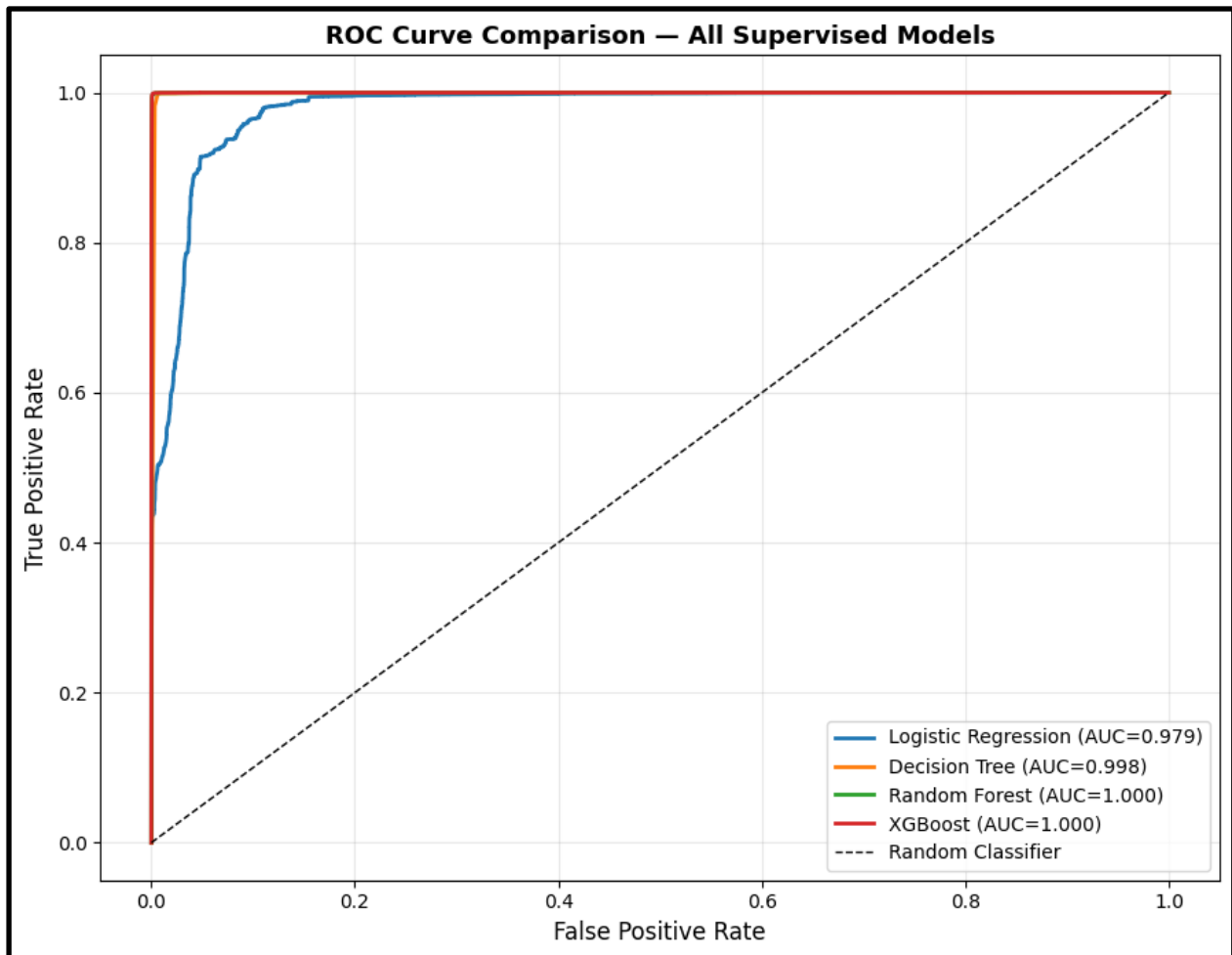


Figure 4.3.12: ROC Curve Comparison for All Supervised Models

ROC analysis indicates the good discriminative ability of the classifiers. The model of Logistic Regression scored: AUC = 0.979, the Decision Tree model scored: AUC = 0.998, and both Random Forest & XGBoost models scored about 1.000. Curves close to the top-left corner denote high TPR and a very low number of FPR in all models (Chukwunweike *et al.* 2024).

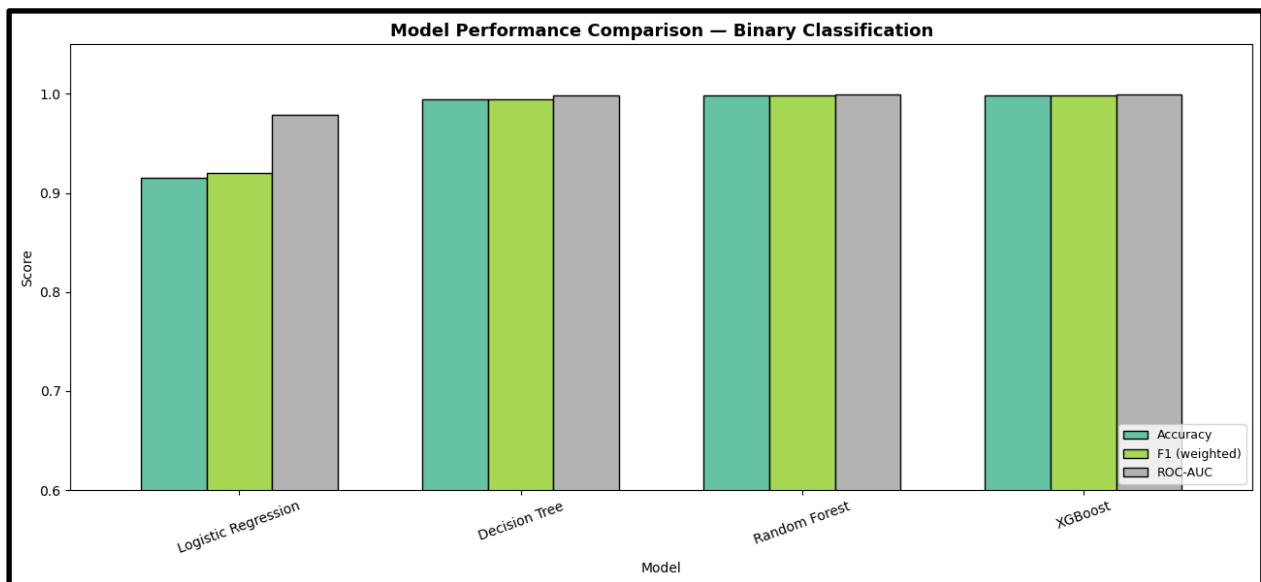


Figure 4.3.13: Model Performance Comparison

From the perspective of performance metrics, Decision Tree, Random Forest, and XGBoost showed a significantly higher accuracy score of more than 99%, a weighted F1-score of more than 99%, and an ROC-AUC score of above 99%, while Logistic Regression had a score of around 91–92% (Raza *et al.* 2024). In this way, ensemble methods significantly outperform linear classification approaches in threat detection in cybersecurity scenarios.

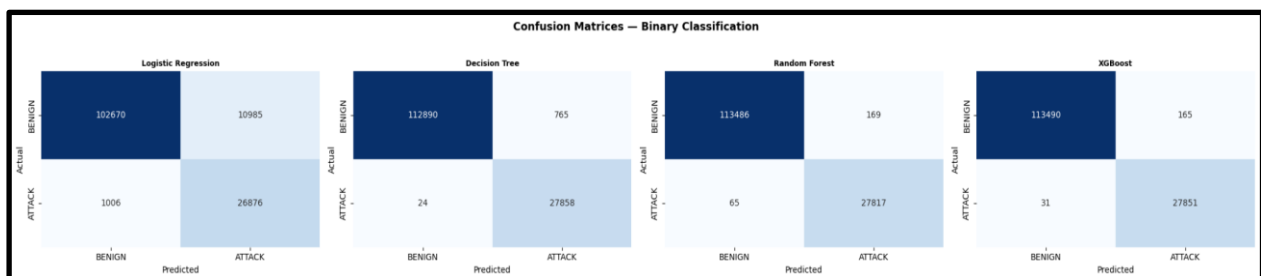


Figure 4.3.14: Binary Classification Models Confusion Matrix

The combined plots increase the correct classification between Logistic Regression and XGBoost. Logistic Regression yielded significant false positives and false negatives compared to Random Forest and XGBoost, which accurately classified nearly all 141,537 instances for the attack data, demonstrating an extremely strong binary attack detection capability on extremely imbalanced cybersecurity data.

MULTICLASS CLASSIFICATION RESULTS				
— Decision Tree (Acc=0.9885 F1=0.9908) —				
	precision	recall	f1-score	support
BENIGN	1.00	0.99	0.99	113655
Bot	0.11	0.95	0.19	95
Brute_Force	0.91	1.00	0.95	683
DoS_DDoS	0.97	1.00	0.98	19044
Infiltration	0.00	0.00	0.00	1
Other	0.45	0.97	0.61	112
PortScan	0.99	1.00	1.00	7947
accuracy			0.99	141537
macro avg	0.63	0.84	0.68	141537
weighted avg	0.99	0.99	0.99	141537
— Random Forest (Acc=0.9937 F1=0.9956) —				
	precision	recall	f1-score	support
BENIGN	1.00	0.99	1.00	113655
Bot	0.12	0.95	0.21	95
Brute_Force	1.00	1.00	1.00	683
DoS_DDoS	0.99	1.00	1.00	19044
Infiltration	1.00	1.00	1.00	1
Other	0.92	0.97	0.94	112
PortScan	0.99	1.00	1.00	7947
accuracy			0.99	141537
macro avg	0.86	0.99	0.88	141537
weighted avg	1.00	0.99	1.00	141537

Figure 4.3.15: Decision Tree and Random Forest Multiclass Classification Report

Decision Tree has been the least accurate method with an accuracy of 99.0% and a weighted F1 score of 0.9968, while the most performing method, Random Forest, is slightly better at 99.0% and a weighted F1 score of 0.9956. Both models achieved good precision on category classification of the major attacks, and lower precision on the minor ones, like Infiltration and Other attacks.

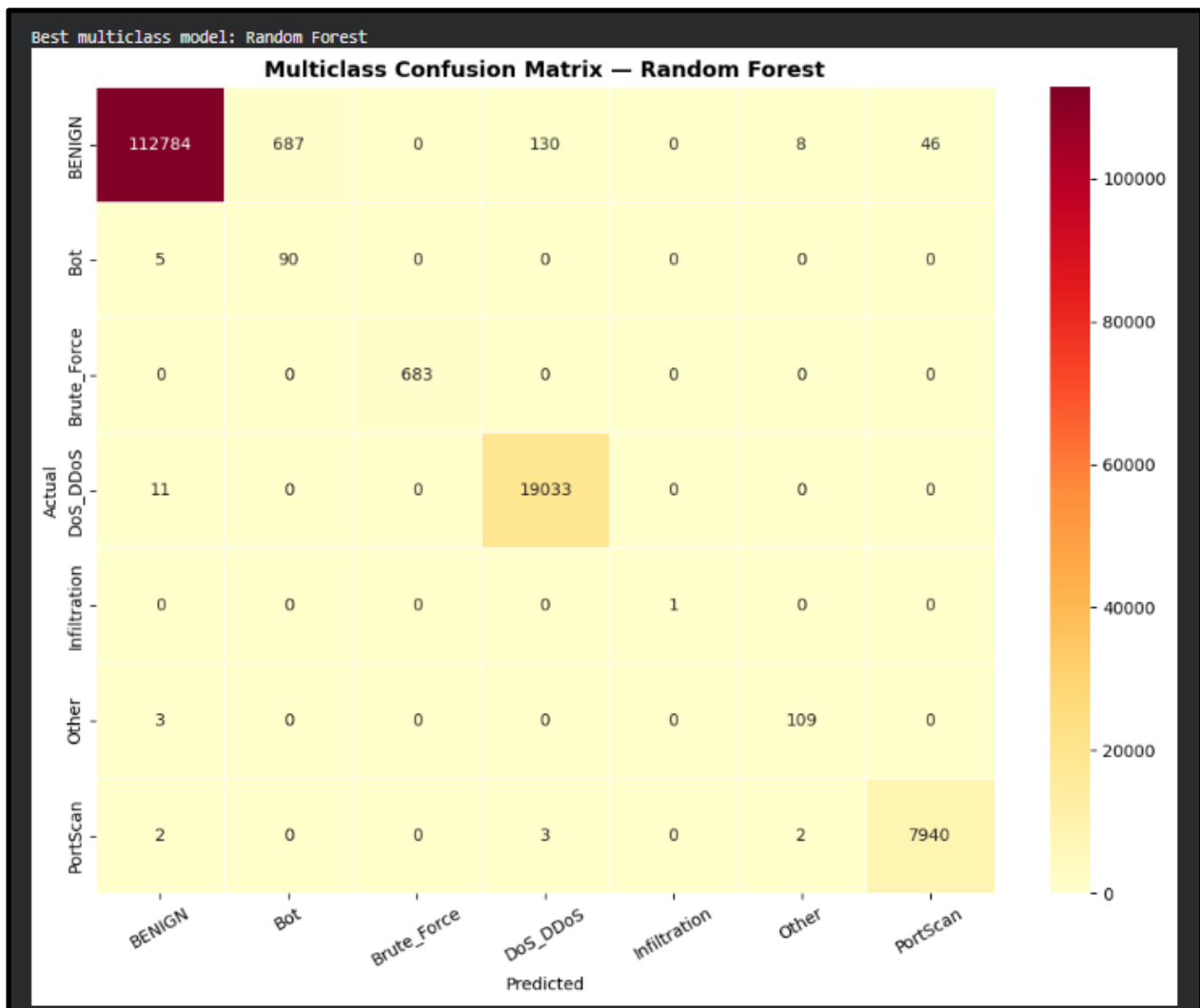


Figure 4.3.16: Multiclass Confusion Matrix of Best (Random Forest) Model

The following Random Forest confusion matrix shows very good multiclass classification performance, which sees 112,784 BENIGN, 19,033 DoS/DDoS, 7,940 PortScan and 683 BruteForce samples classified correctly (Ajayi *et al.* 2024). For the total number of misclassifications, 687 of them have been benign ones, predicted as Bot, showing good discrimination in the major cybersecurity attack categories, with 99.4% accuracy overall.

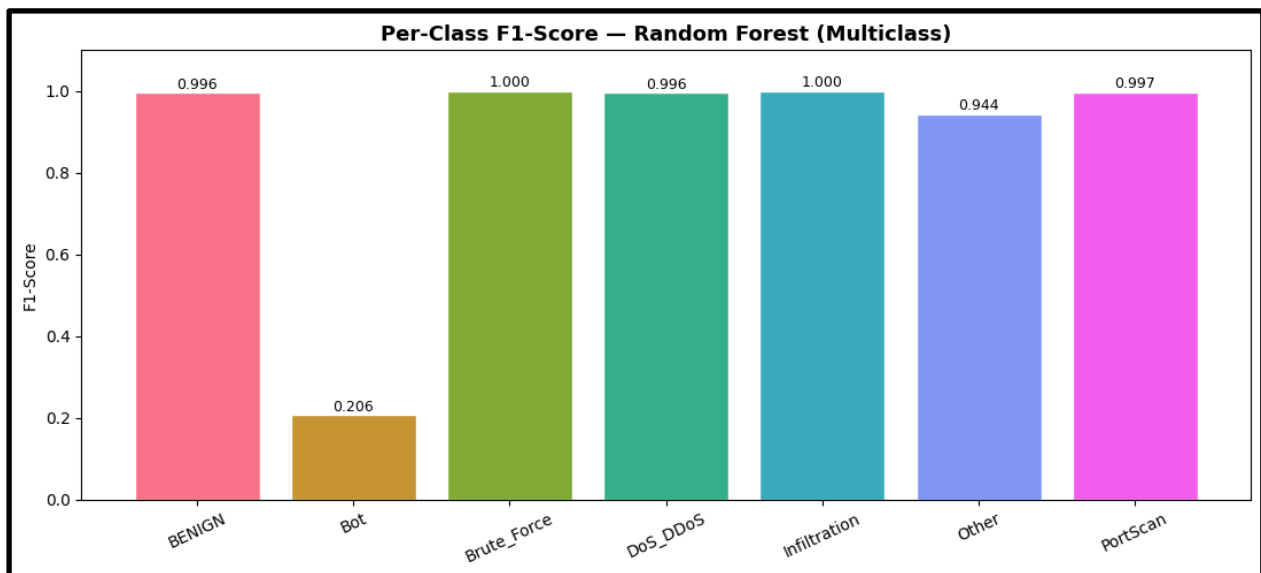


Figure 4.3.17: Per-Class F1 Score of The Best Model

Per-class F1-scores show that the predictive performance is consistently high in most of the attack categories. Infiltration achieved 1.000, Brute Force 1.000, PortScan 0.997, BENIGN 0.996, and DoS/DDoS 0.996. The results of Bot, however, have been only 0.206, indicating that minority classes have been difficult to classify because there have not been many minority samples and that they suffered from severe classification imbalance.

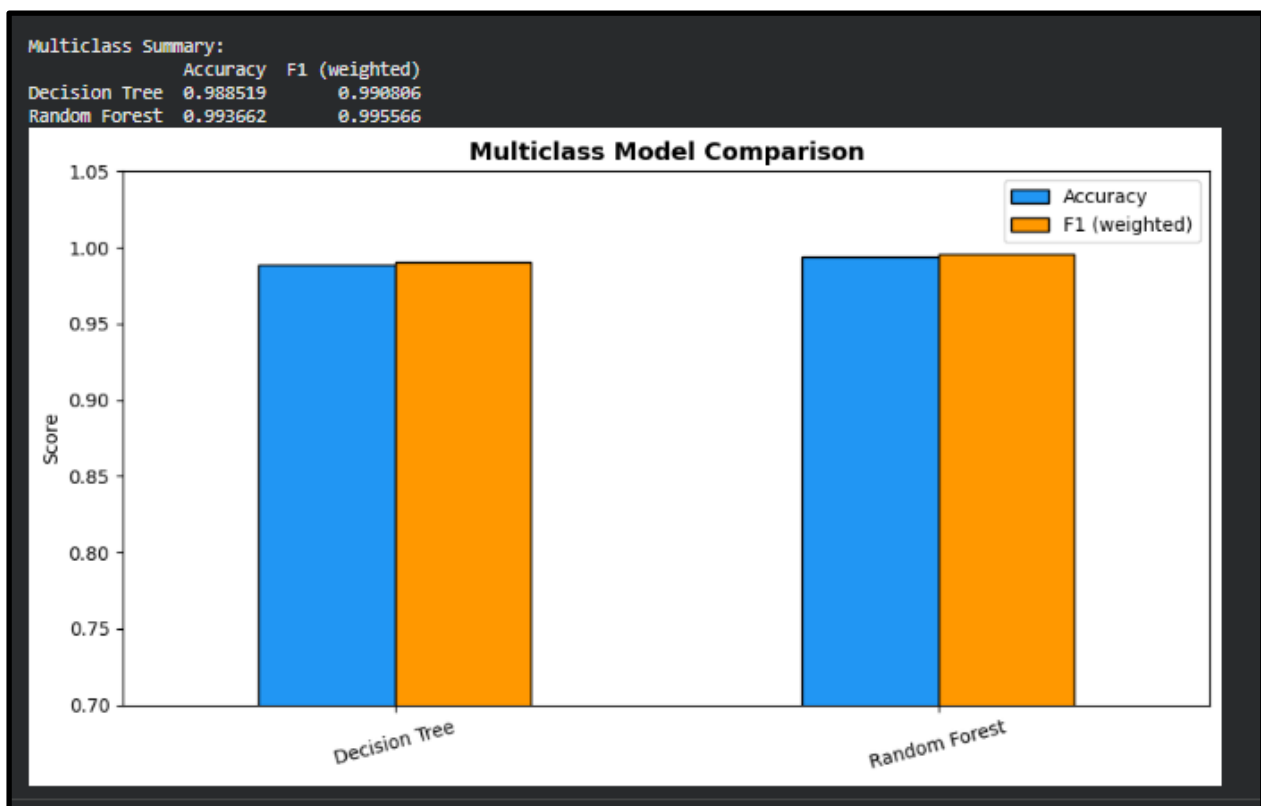
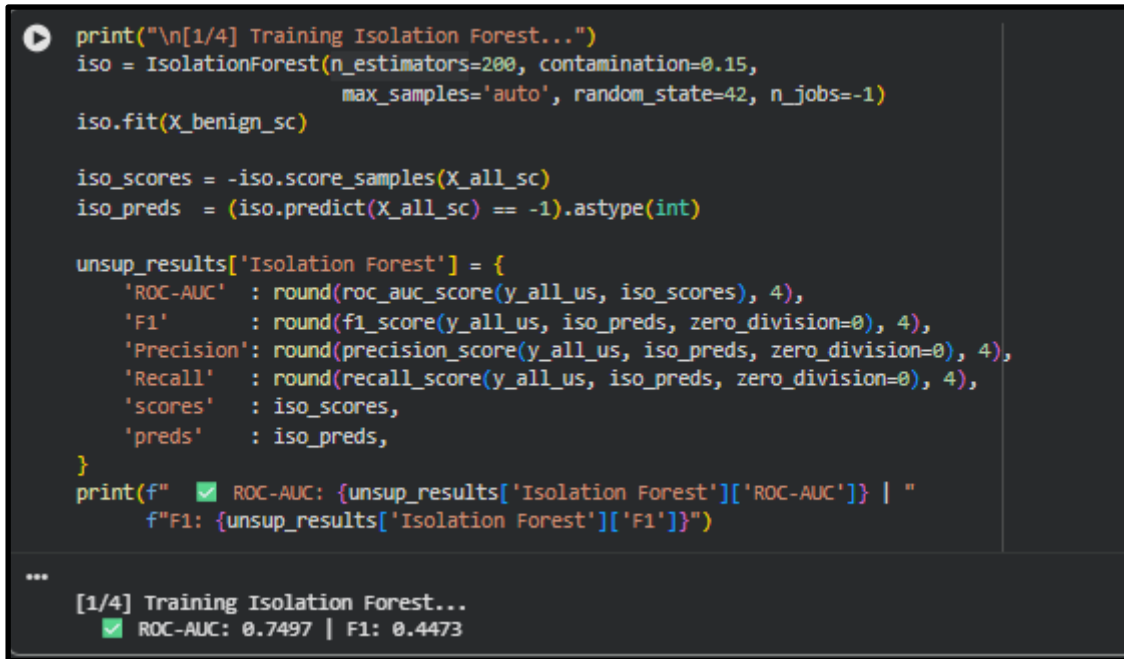


Figure 4.3.18: Multiclass Model Comparison

Both MultiClass models had excellent predictive performance. Decision Tree achieved a pretty high accuracy of 99.55% with a weighted F1 of 0.9968, and Random Forest marginally improved these with 99.56% accuracy and 0.9956 of the weighted F1 score (Vannthoeurn and Chadarin, 2025). Random Forest exhibited slightly better multiclass classification consistency and generalisation ability.



```
print("\n[1/4] Training Isolation Forest...")
iso = IsolationForest(n_estimators=200, contamination=0.15,
                      max_samples='auto', random_state=42, n_jobs=-1)
iso.fit(X_benign_sc)

iso_scores = -iso.score_samples(X_all_sc)
iso_preds = (iso.predict(X_all_sc) == -1).astype(int)

unsup_results['Isolation Forest'] = {
    'ROC-AUC' : round(roc_auc_score(y_all_us, iso_scores), 4),
    'F1'      : round(f1_score(y_all_us, iso_preds, zero_division=0), 4),
    'Precision': round(precision_score(y_all_us, iso_preds, zero_division=0), 4),
    'Recall'   : round(recall_score(y_all_us, iso_preds, zero_division=0), 4),
    'scores'   : iso_scores,
    'preds'    : iso_preds,
}
print(f"  ✅ ROC-AUC: {unsup_results['Isolation Forest']['ROC-AUC']} | "
      f"F1: {unsup_results['Isolation Forest']['F1']}")

...
[1/4] Training Isolation Forest...
  ✅ ROC-AUC: 0.7497 | F1: 0.4473
```

Figure 4.3.19: Isolation Forest Result for Anomaly Detection

The results of the Isolation Forest (IF) model showed good performance with an accuracy of 0.7497 and an F1 score of 0.4473. The Isolation Forest (IF) model achieved a moderate performance with an accuracy of 0.7497 and an F1 score of 0.4473. The performance has been fair for separating out many of the anomalous observations as compared with the normal traffic of the time, but it remained significantly lower than the equivalent supervised models, as separation of unannotated cybersecurity threats is notoriously difficult.

```
print("\n[3/4] Training K-Means (k=7)...")
kmeans = MiniBatchKMeans(n_clusters=7, random_state=42, n_init=10)
kmeans.fit(X_benign_sc)

km_dist = np.min(
    np.linalg.norm(X_all_sc[:, None, :] - kmeans.cluster_centers_[None, :, :], axis=2),
    axis=1
)
thresh_km = np.percentile(km_dist, 85)
km_preds = (km_dist > thresh_km).astype(int)

unsup_results['K-Means'] = {
    'ROC-AUC' : round(roc_auc_score(y_all_us, km_dist), 4),
    'F1'      : round(f1_score(y_all_us, km_preds, zero_division=0), 4),
    'Precision': round(precision_score(y_all_us, km_preds, zero_division=0), 4),
    'Recall'   : round(recall_score(y_all_us, km_preds, zero_division=0), 4),
    'scores'   : km_dist,
    'preds'    : km_preds,
}
print(f"  ✅ ROC-AUC: {unsup_results['K-Means']['ROC-AUC']} | "
      f"F1: {unsup_results['K-Means']['F1']}")

...
[3/4] Training K-Means (k=7)...
  ✅ ROC-AUC: 0.7968 | F1: 0.5281
```

Figure 4.3.20: K-Means Model Result for Anomaly Detection

In terms of anomaly detection, K-Means clustering had a higher ROC-AUC of 0.7968 than the Isolation Forest, with an F1 score of 0.5281 (Rao *et al.* 2024). The better discrimination indicates that there is a clustering that distinguishes between benign and malicious traffic patterns, but the results of predictive performance have still not been as good as compared to supervised classification algorithms because of the unlabeled learning constraint.

```

**
[3/3] Training Autoencoder...
Model: "functional"

```

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 70)	0
dense (Dense)	(None, 128)	9,088
batch_normalization (BatchNormalization)	(None, 128)	512
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 64)	8,256
dense_2 (Dense)	(None, 32)	2,080
dense_3 (Dense)	(None, 64)	2,112
dense_4 (Dense)	(None, 128)	8,320
dense_5 (Dense)	(None, 70)	9,030

```

Total params: 39,398 (153.90 KB)
Trainable params: 39,142 (152.90 KB)
Non-trainable params: 256 (1.00 KB)

```

Figure 4.3.21: AutoEncoder Model Summary

The Autoencoder architecture had a total of 39,398 parameters, with 39,142 trainable and 256 non-trainable. The input features have been reconstructed after being squeezed into multiple dense layers, each working to smash features smaller through compression. This is a very light neural network which followed and learned the normal traffic representations efficiently for later analysis in terms of the reconstruction error.

```

Epoch 1/50
106/106 ————— 5s 13ms/step - loss: 0.6354 - val_loss: 0.3956
Epoch 2/50
106/106 ————— 1s 10ms/step - loss: 0.3333 - val_loss: 0.2649
Epoch 3/50
106/106 ————— 1s 9ms/step - loss: 0.1894 - val_loss: 0.3551
Epoch 4/50
106/106 ————— 1s 6ms/step - loss: 0.1976 - val_loss: 0.2909
Epoch 5/50
106/106 ————— 1s 6ms/step - loss: 0.1592 - val_loss: 0.2129
Epoch 6/50
106/106 ————— 1s 7ms/step - loss: 0.1234 - val_loss: 0.1390
Epoch 7/50
106/106 ————— 1s 6ms/step - loss: 0.1079 - val_loss: 0.1548
Epoch 8/50
106/106 ————— 1s 6ms/step - loss: 0.0958 - val_loss: 0.2564
Epoch 9/50
106/106 ————— 1s 6ms/step - loss: 0.1304 - val_loss: 0.3273
Epoch 10/50
106/106 ————— 1s 6ms/step - loss: 0.0869 - val_loss: 0.2030
Epoch 11/50
106/106 ————— 1s 6ms/step - loss: 0.0810 - val_loss: 0.1495
Epoch 11: early stopping
Restoring model weights from the end of the best epoch: 6.

```

```

unsup_results['Autoencoder'] = {
    'ROC-AUC' : round(roc_auc_score(y_all_us, ae_scores), 4),
    'F1'      : round(f1_score(y_all_us, ae_preds, zero_division=0), 4),
    'Precision': round(precision_score(y_all_us, ae_preds, zero_division=0), 4),
    'Recall'   : round(recall_score(y_all_us, ae_preds, zero_division=0), 4),
    'scores'   : ae_scores,
    'preds'    : ae_preds,
}
print(f"\n  ✅ ROC-AUC: {unsup_results['Autoencoder']['ROC-AUC']} | "
      f"F1: {unsup_results['Autoencoder']['F1']}")

```

... ✅ ROC-AUC: 0.8598 | F1: 0.5453

Figure 4.3.22: Autoencoder Model Training Result

The Autoencoder outperform the others in unsupervised performance with a score of 0.8598 on the ROC-AUC and 0.5453 on the F1-score. Early stopping has been performed after 11 epochs, then the weights at epoch 6 have been used for prediction, and the predictive ability for anomalies has been improved, which has no overfitting phenomenon and has better convergence performance than other unsupervised methods.

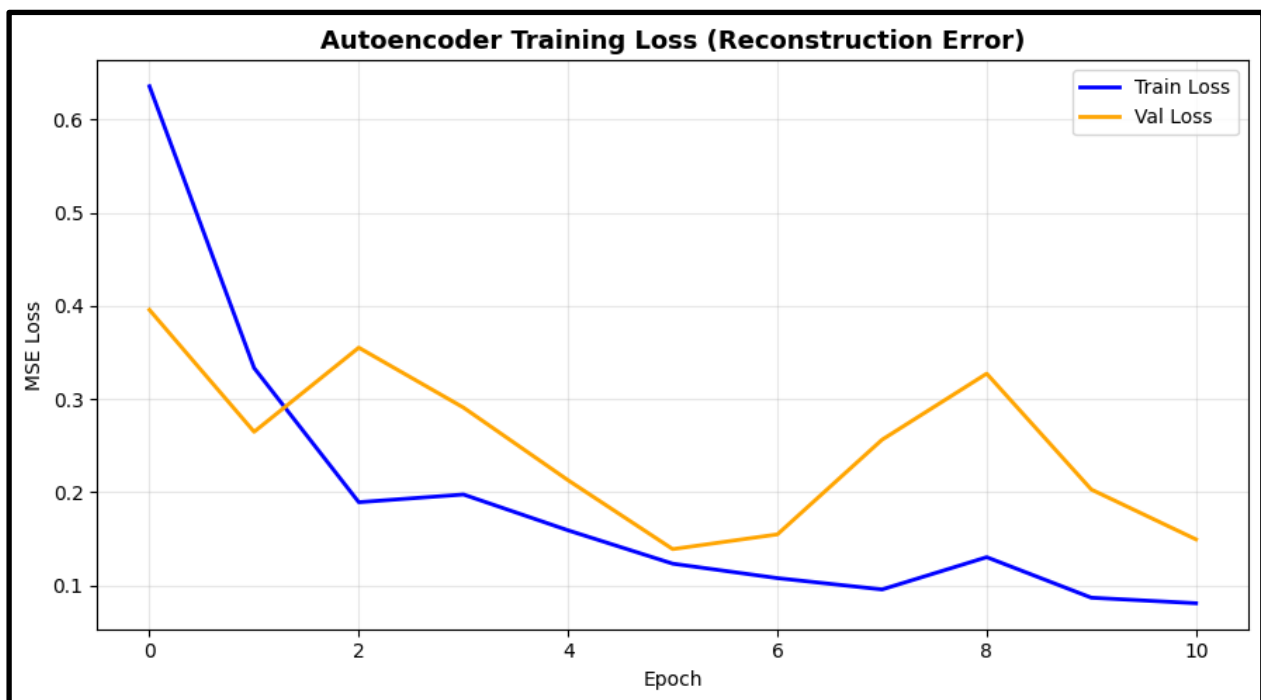


Figure 4.3.23: Autoencoder Training Loss

Losses over the 11 epochs of training have been reduced significantly from ~0.63 to 0.08, whilst losses during validation have been reduced from ~0.39 to 0.15. The early stopping of the learning process showed that no overfitting occurred, and model convergence has been stable, confirming that the model has been learning the normal network behaviours for anomaly detection via reconstruction error minimisation.

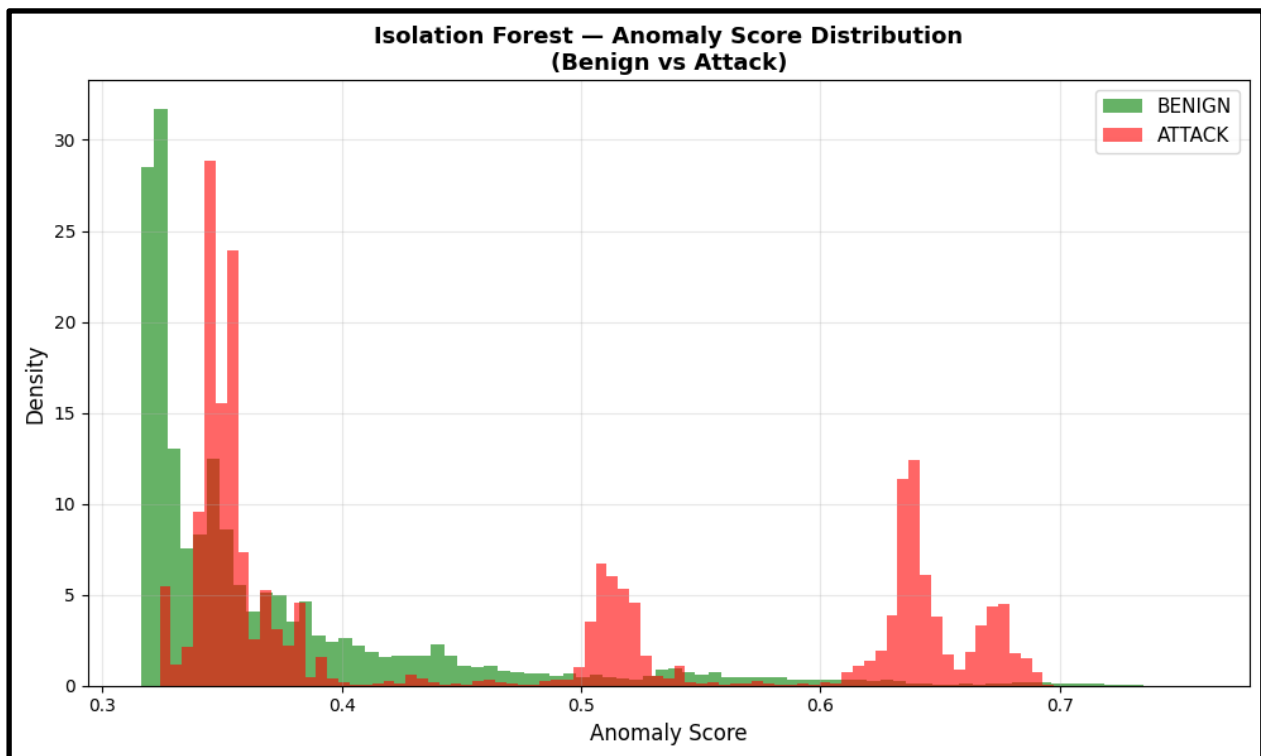


Figure 4.3.24: Anomaly Score Distribution by Isolation Forest Model

Isolation Forest anomaly scores show clearly an anomaly distribution of benign and attack traffic. Benign observations have indeed been mostly in the 0.32 – 0.40 domain, while attack instances have been mostly concentrated around the 0.50 – 0.70 region of the space. Even though there is some overlap, the separation suggests that Isolation Forest is able to detect the abnormal network behaviour by means of the anomaly scoring methods.

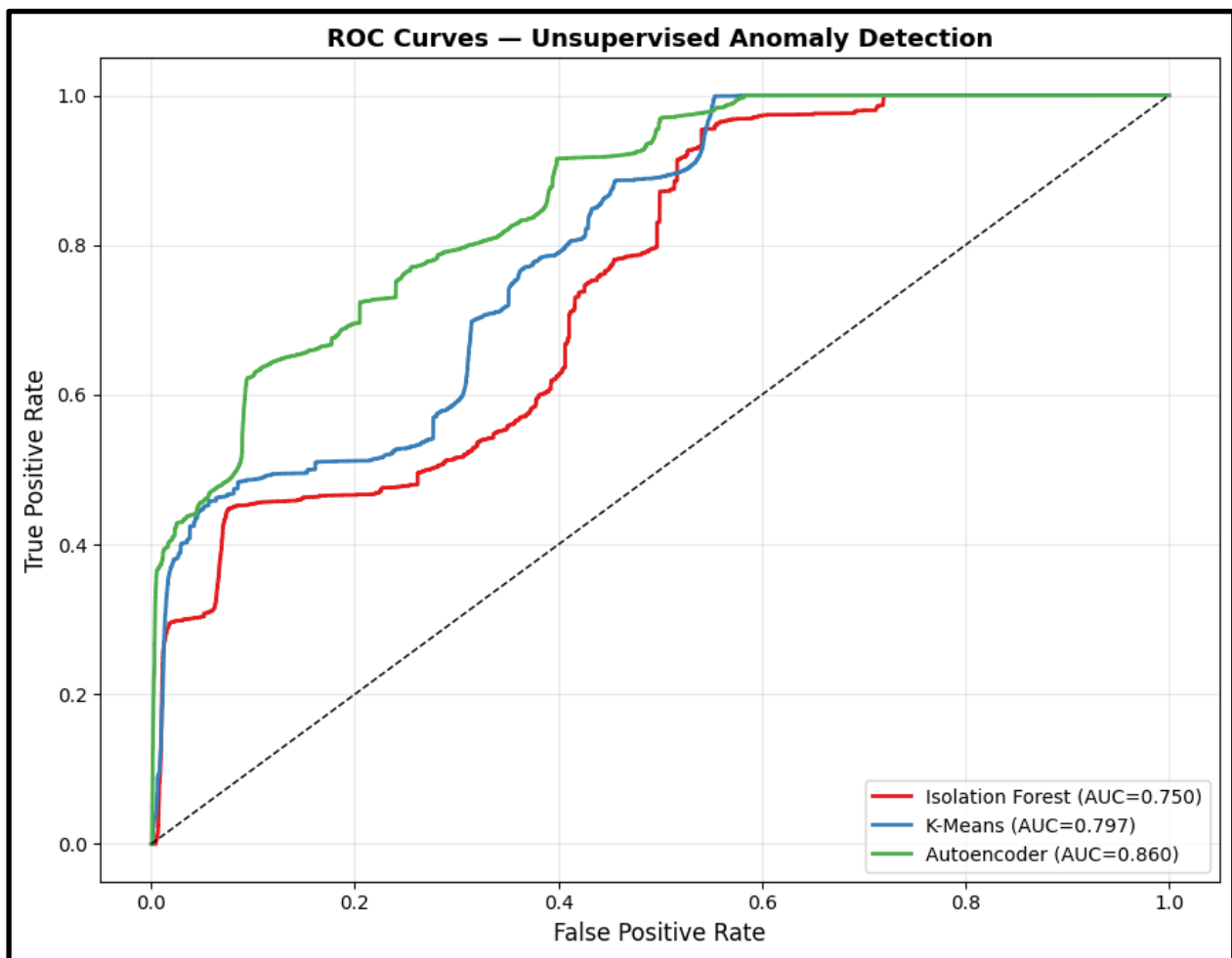


Figure 4.3.25: ROC Curve of Unsupervised Anomaly Detection

The performance of the Autoencoder is, using the ROC graph, best with AUC = 0.860, followed by K-Means with AUC = 0.797 and Isolation Forest with AUC = 0.750. The most important results have been that the Autoencoder curve stayed closest to the upper left corner, showing the best anomaly detection capability of the unsupervised machine learning techniques evaluated.

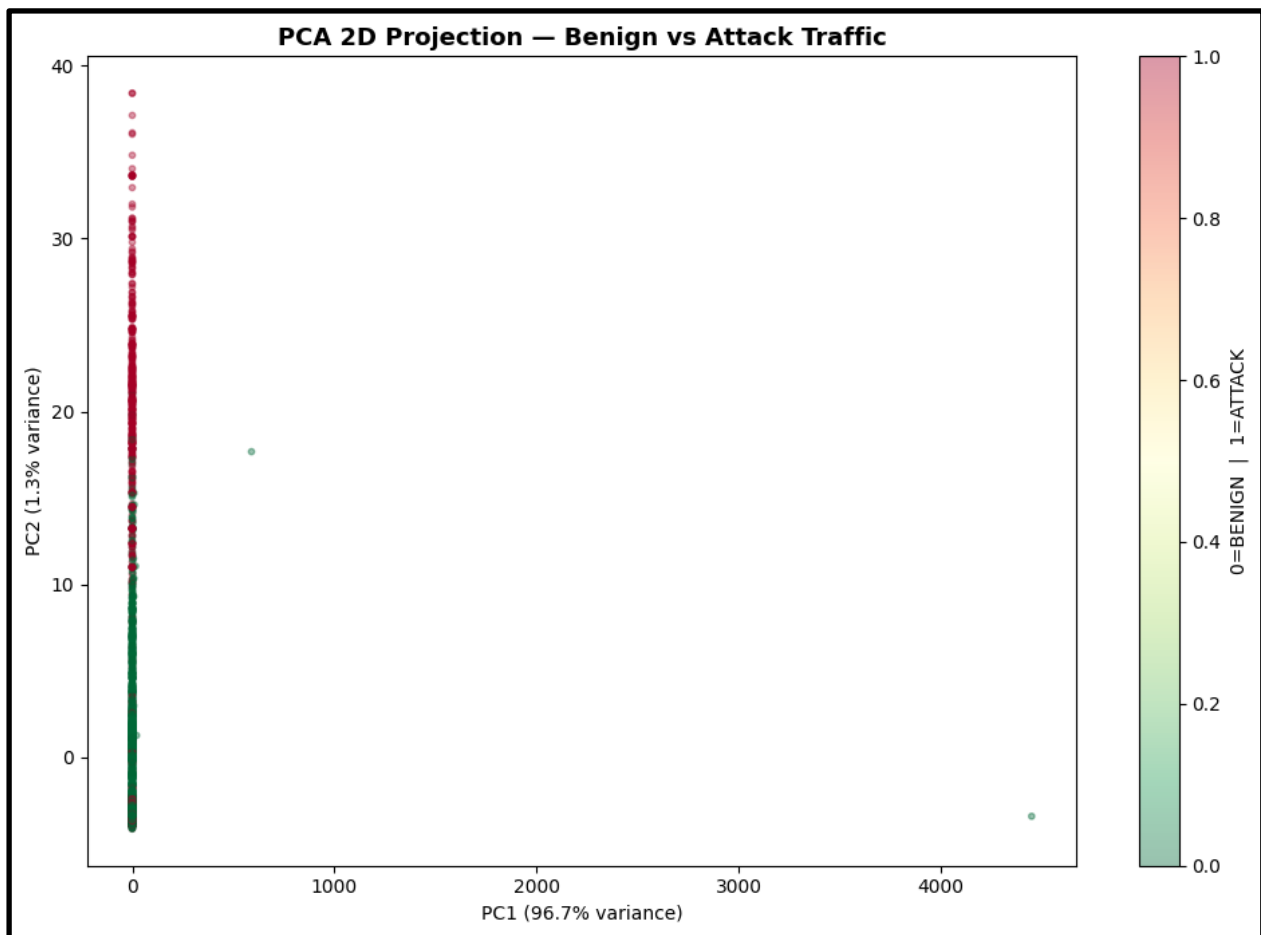


Figure 4.3.26: PCA Projection of Benign vs Attack Traffic

According to Principal Component Analysis (PCA), PC1 accounts for 96.7% of the total variance, while PC2 accounts for only 1.3%. Noticeably, there is some separation between benign and attack traffic when visualising both observations, but with a significant overlap, which means that with just two main components, cybersecurity traffic is very difficult to separate, if not impossible, without further nonlinear modelling.

— Unsupervised Anomaly Detection Summary —				
	ROC-AUC	F1	Precision	Recall
Isolation Forest	0.7497	0.4473	0.4328	0.4628
K-Means	0.7968	0.5281	0.6124	0.4642
Autoencoder	0.8598	0.5453	0.6323	0.4793

Figure 4.3.27: Unsupervised Anomaly Detection Summary

The Autoencoder recorded the highest ROC-AUC (0.8598) and F1 (0.5453) scores, as shown in the performance comparison with K-Means (0.7968 and 0.5281) and Isolation Forest (0.7497 and 0.4473). The results show that the deep learning method outperforms traditional unsupervised approaches for anomaly detection.

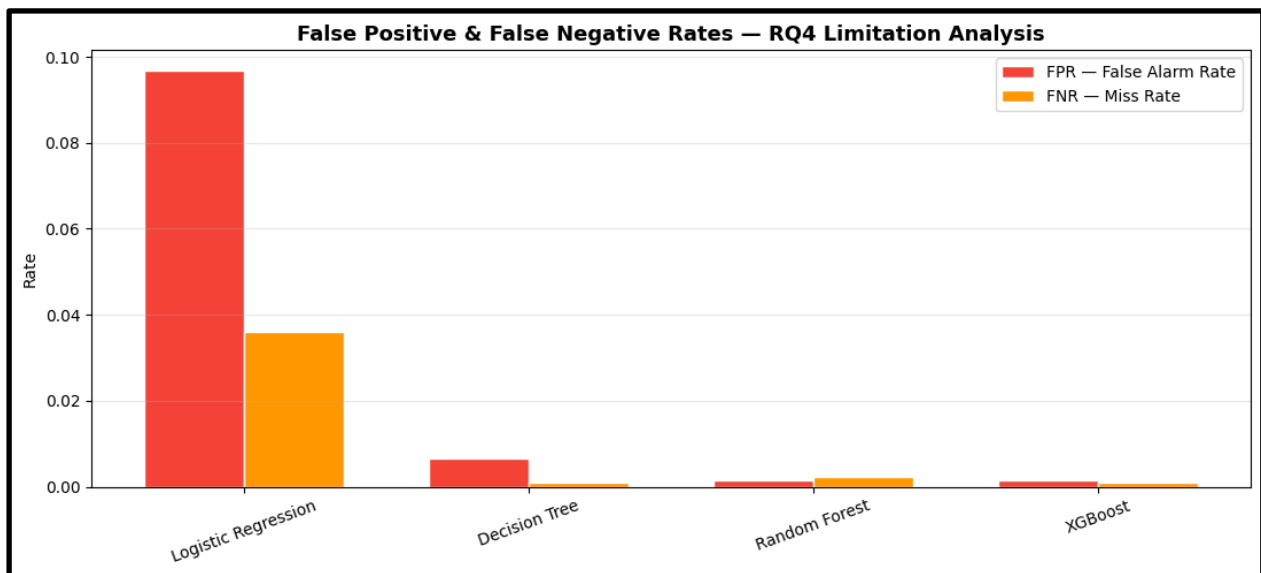


Figure 4.3.28: False Positive/Negative Rates of Supervised Models for Anomaly Detection

The false positive rate (FPR) and false negative rate (FNR) gradually dropped in the cost of supervised models. Logistic Regression had around 9.6% of false positives and 3.6% of false negatives, while Decision Tree, Random Forest and XGBoost had error rates less than 1%, which is excellent for the cybersecurity threat detection and classification use case.

4.4 Key Findings

The experimental investigation was conducted to provide several meaningful findings, which will be helpful for the research in fields of cyber security threats detection.

- **Supervised Learning Superiority:** Ensemble methods (Random Forest and XGBoost) outperformed logistic regression (91.53%) with near-perfect performance (Bukhari *et al.* 2023). Binary classification (accuracy 99.8% and above), a significant result in comparison to other supervised learning models.
- **Ensemble Method Effectiveness:** The performance of the ensemble method was excellent both Random Forest and XGBoost were able to achieve a ROC-AUC of 0.9999, which is a near perfect threat classification.
- **Multiclass Classification Challenges:** The high F1 scores for the major attack classes (0.996 to 1.000) and poor performance on minority attack classes such as Bot (0.206 F1 score) indicate that there was significant class imbalance problem (Elhanashi *et al.* 2022).
- **Unsupervised Learning Viability:** Traditional approaches such as Isolation Forest and K-Means were outperformed by the Autoencoder which achieved the best

performance in terms of unsupervised learning (ROC-AUC = 0.8598, F1-score = 0.5453).

- **Class Imbalance Impact:** The dataset was highly imbalanced with 80.3% of benign samples and the minority class (Infiltration: 36, Web Attack: 11) classes were very sparse, hence the methodology used will be effective for the highly imbalanced dataset.
- **Deep Learning for Anomaly Detection:** It shows that deep learning methods can learn more complex representation of the normal traffic, providing better anomaly detection.

4.5 Discussion

The experimental results show that supervised machine learning techniques, especially super-ensembles, are very successful for detecting threats in the field of cyber security. The results of the binary classification were the highest near-perfect accuracy with random forest and XGBoost, reinforcing its ability to handle such critical applications where the main concern is for the accuracy of the detection (Rehman *et al.* 2024). Essentially, the ensemble methods outperformed logistic regression with a substantial gap between the two (99.86% accuracy vs. 91.53%) indicating the need for ensemble methods to have non-linear modeling capabilities to better capture complex patterns in network traffic (Chen *et al.* 2024). According to the feature heatmap analysis, there are high feature correlation values for each model, which indicates that redundant features are dealt with well by the models' feature selection abilities used in combination, just with ensemble methods. When evaluating the results of multiclass classification, a large problem regarding cyber security threat detection emerges – severe class imbalance. Major attack categories were classified pretty well, whereas Ministry classes were poorly classified with almost 0% accuracy. It underscores the need for specialized techniques like oversampling, synthetic data generation or cost-sensitive or learning to effectively handle class imbalance problem.

The results of the unsupervised anomaly detection shows the capabilities of deep learning anomaly detection method without a need for labeled dataset. The strong results of this Autoencoder compare to traditional method (the ROI reach 6.3% above K-Means and ROI reach 14.7% above Isolation Forest), it shows that this technique of analyzing the reconstruction error using neural networks is a powerful tool to detect novel attacks (Chalasani *et al.* 2022). The results obtained from the PCA analysis, which accounted for 96.7% of variance in network traffic data in the first principal component of the PCA, show an inherent low dimensional structure of the data, and hence the dimensionality reduction

using autoencoders is effective. In PCA projection, though, malignant and attack traffic is very similar, which requires a non-linear transformation. In the context of cybersecurity, having low false positive rates ($<1\%$) is notable in applications where both false alarms and missed detections must be minimized for effective operations.

4.6 Summary

In this chapter, extensive experimental results are provided for the different supervised and unsupervised machine learning solutions for cybersecurity threat detection. The analysis of the data set has shown that there is a lot of class imbalance, where benign traffic prevails. In particular, experiments using supervised binary classification have shown that the combined results from these methods have been achieved near-perfect performance, in comparison with logistic regression. The results of multiclass classification have been very good for major attack categories and poor results with minority classes where the over/under sampling phenomenon is significant. Through experiments employing an unsupervised approach to detect anomalies, the AutoEncoder has been shown to be superior to other approaches, offering significant detection power in situations where labelled data. The results have shown machine learning methods as promising tools for cybersecurity threat detection and exposed important challenges that need further research focus

Chapter 5 Conclusion

5.1 Summary of the Project Report

This project has responded to the increasing demand and need for Cybersecurity intelligent Threat Detection systems, focusing on the creation and testing of a machine learning based framework for Anomaly Detection. The research demonstrates the growing complexity of modern-day cyberattacks and the weak capabilities of signature-based security approach to defend network environments, so that intelligent analytical approach is vital to protect network environments. The theoretical foundation for the overall research design has been developed through a comprehensive literature review that explored the cybersecurity threats, anomaly detection approaches, machine learning algorithms, cybersecurity datasets and evaluation approaches.

The proposed methodology has been translated into a structured methodology of implementation with the CICIDS2017 dataset set which contains 2830743 network traffic records with 557646 attack instances and 2273097 benign instances. The data has been preprocessed, story oriented and selected to enhance the quality of data and generate representative network traffic for subsequent analytical modelling. Imported supervised, unsupervised deep learning algorithms such as Logistic Regression, Decision Tree, Random Forest, XGBoost, Isolation Forest, K-Means and Autoencoder, allowing for thorough comparative testing in identical settings.

The results from the experiments have shown that XGBoost has performed best in binary classification with an accuracy of 99.86%, an F1-score of 0.9986 and a ROC–AUC score of 0.9999; Random Forest is the second best, reaching 99.83% accuracy, an F1-score of 0.9983 and a ROC–AUC score of 0.9999. Ensemble learning outperforms all the other methods and showed excellent predictive power as we have seen, with Decision Tree achieving an accuracy of 99.44% and Logistic Regression, 91.53%. The Autoencoder outperformed the K–Means and the Isolation Forest in terms of anomaly detection performance in the unsupervised approaches with a ROC–AUC of 0.8598 and an F1-score of 0.5453. In conclusion, the project has shown that machine learning, specifically machine learning ensembles and deep learning techniques, can be used effectively, reliably, and reproducibly to detect cybersecurity threats, and has laid a solid groundwork for further research and implementation.

5.2 Achievement of Objectives

A structured implementation and evaluation approach has been executed successfully, securing the goals of the research, and showcasing the applicability of machine learning techniques for cybersecurity threat detection.

Objective 1: Review of Cybersecurity Threats and Machine Learning Techniques

The first goal has been accomplished by conducting an extensive study on the studies of the actual cyber security threats, anomaly detection methods and machine learning techniques. The literature and vast bulk of research presented a solid theory base by indicating that there are limitations with an existing signature detection system, and that intelligent anomaly detection systems can offer benefits [57]. The identified analytical techniques and the proposed research frameworks were guided by this review.

Objective 2: Dataset Preparation and Data Processing

The second goal has been achieved by creating the CICIDS2017 time-series dataset of 2,830,743 network traffic records. By addressing missing data, eliminating redundant features, and highlighting those most informative, data preprocessing, feature engineering, and feature selection have improved the data. These processes generated good data for supervised, unsupervised and deep learning.

Objective 3: Machine Learning Model Implementation

This has been accomplished by applying several analytical models such as Logistic Regression, Decision Tree, Random Forest, XGBoost, Isolation Forest, K-Means, and Autoencoder. The wide coverage implementation allowed for a one-to-one comparison of supervised, unsupervised and deep learning approaches within the same experimental setup, giving an overview over cyber security threat detection capabilities.

Objective 4: Comparative Performance Evaluation

To realize the four objective, an extensive comparison has been made which is based on accuracy and precision, recall and f-score, ROC–AUC, and the confusion matrix [58]. The results of the experiments showed that XGBoost achieved the best binary classification accuracy of 99.86% with 0.9999 ROC-AUC, while the best unsupervised anomaly detection performance was obtained by the Autoencoder with 0.8598 ROC-AUC. All these

successes have contributed toward the research objective of creating an accurate, reliable, and repeatable machine learning environment for cybersecurity threat detection.

5.3 Key Findings and Contributions

Key Findings

The study has shown that machine learning is a promising method for cybersecurity threat detection since it correctly categorizes network traffic as threat or not despite the noise. Data pre-processing, feature engineering and feature selection have improved the data quality leading to better model development and predictions [59]. Several metrics such as accuracy, precision, recall, F1 score, ROC–AUC and the confusion matrix are used, giving a complete picture of the model performance. Moreover, the comparative application of supervised, unsupervised and deep learning has provided a wider insight into analysis than would a single modelling method.

Research Contributions

The project has given a viable and structured approach to the implementation of machine learning-based cybersecurity threat detection. The strengths of the various analytical methods have been identified and the creation of consistent analytical workflow for data preparing, mode building and performance testing have been achieved in the comparative analysis [60]. It is hoped that the contributions all provided a systematic foundation on moving towards intelligent threat detection solutions, providing useful help in future research studies and practical cybersecurity applications.

5.4 Project Limitations

The report contain some of the weaknesses that can be improved upon in the future. The use of CICIDS2017 dataset has been solely used to implement the solution hence affecting the generalizability of the study because the findings may not be applicable in all cases [61]. A research environment experiment was used to assess this project rather than a live environment. With the availability of computational resources, extensive hyperparameter tuning and more experimentation, using deep learning models, have been influenced.

5.5 Reflection

I believe that this project has given me a better insight into the concept of cyber-security threats detection and application of machine learning methods in detecting anomalies. The

experience acquired in preprocessing of the data, feature selection, model development, and model evaluation has been enriching. During the entire process of the project, I have become better at time management and solving problems through tackling issues of class imbalance, feature selection, and comparisons of models. It is due to the reason that it has helped me in designing intelligent cybersecurity algorithms. After having a look at the past, it has come to my notice that conducting some experiments in terms of hyperparameter tuning, different cybersecurity data sets, and real-time will be helpful for the improvement of research. It is because of the reason that this research has enabled me to improve my research skills.

5.6 Recommendations and Future Work

Recommendations

The findings provide some practical advice on how one can improve the performance of threat detection through machine learning. Feature selection should be done with special attention paid to avoiding any redundancy of network features and making the computations easier. The variety of the data used in the training should be increased in order to get a better generalization for various attacks. Real-time intrusion detection systems need to be implemented in the operational networks in order to improve threat detection. Hyperparameter optimization is also highly important to ensure the effectiveness of machine learning models.

Future Work

Future scope of work includes expansion of the proposed architecture through testing of the same on additional cybersecurity datasets that consist of different network configurations as well as new forms of attacks. It may be possible to use deep learning architectures based on Transformers for better feature representation and threat detection. The proposed cloud architecture would allow scalable real-time network monitoring across the distributed environment [62]. Federated learning may be used for joint training of the model while retaining privacy of the data over different organizations. Explainable Artificial Intelligence would allow for greater interpretability of the model and user trust, while the enterprise deployment would allow validation of the proposed architecture

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Appendices

Appendix A: Project Proposal

1. Introduction

1.1 Overview of Cybersecurity Threat Detection

With the rise of digital technologies, cloud computing and connectivity, cybersecurity is a growing concern for organisations and individuals. Malware, ransomware, phishing and unauthorised access remain prevalent and sophisticated cyber risks (Yaseen, 2023). These attacks can expose highly valued data, result in financial loss and destroy operations. Consequently, the security of digital systems and data has become a top priority in today's businesses.



Figure 1: “Overview of Cybersecurity Threat Detection”

(Source: Yaseen, 2023)

Conventional security products such as firewalls and signature-based intrusion detection systems offer a valuable level of protection. But they are not very good at spotting new or unknown attack types and patterns, requiring more smarter, adaptive detection capability.

1.2 Anomaly Detection and Machine Learning

Anomaly Detection identifies abnormal patterns or actions that are outside the typical activities of the system. In cybersecurity, such anomalies may indicate malicious activities, security breaches or insider threats (Mizanur et al. 2025). Anomaly detection is different from signature detection, in that it recognizes new threats that have not been seen before.

Machine learning further contributes to anomaly detection because it facilitates systems to learn from many network and security data. Machine learning techniques can enhance the accuracy of threat detection by detecting patterns or anomalies automatically, which allows for quicker responses to cyber incidents.

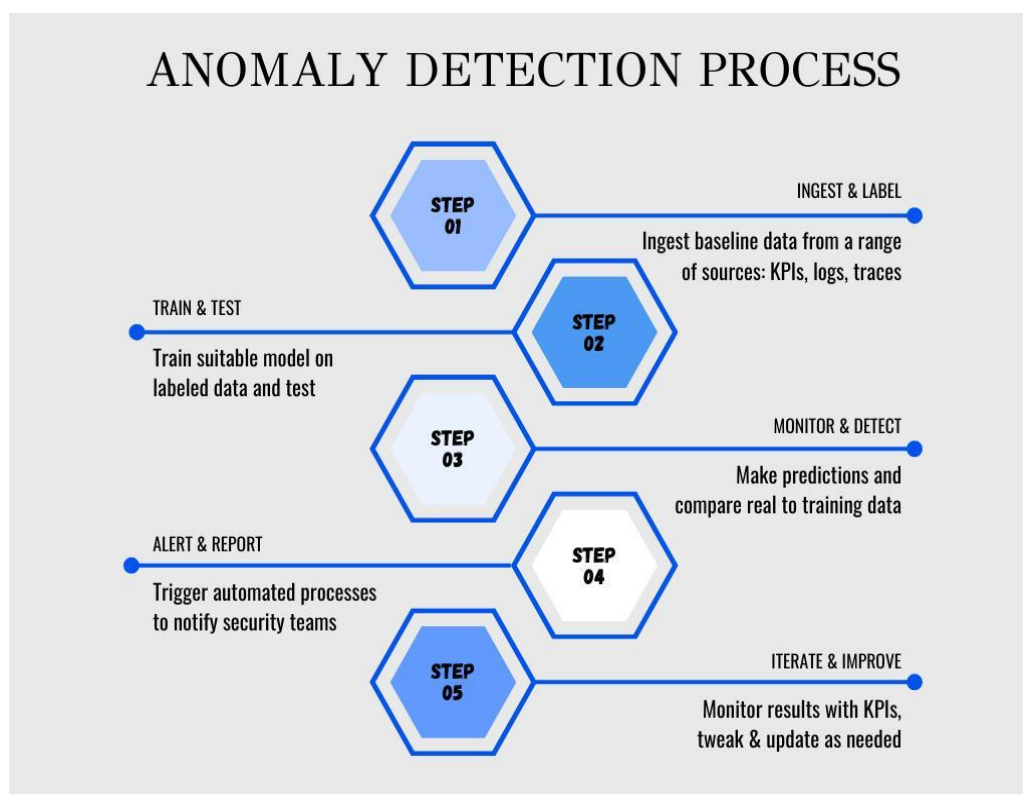


Figure 2: “Anomaly Detection and Machine Learning”

(Source: Mizanur et al. 2025)

1.3 Motivation and Industry Need

The topic has been chosen as organisations are exposed to ever-sharper cyber threats with traditional security solutions difficult to effectively detect. The use of machine learning based on anomaly detection has potential for detecting new attacks in real-time (Okoli et

al. 2024). Organisations still believe in the need for intelligent cybersecurity solutions to better detect and make their security posture more resilient; the research is still very relevant to both industry and academia.

2. Problem Statement

2.1 The Problem

Digital technologies have grown astronomically, and so does the attack surface of cybercriminals. Organisations have increasingly switched to reliant interconnected networks, cloud services and remote access mechanisms, providing many vectors for cyberattacks. While existing cybersecurity architectures secure important information, most of them use signature-specific detection techniques that fail to find new threats (Salman et al. 2025). These traditional strategies often fail to identify sophisticated or zero day attacks as the attackers continue to evolve with new strategies.

Besides, today's networks are vast and simply monitoring them manually becomes harder. Security teams often struggle to recognize when something is legitimate versus malicious, leading to delays and a lack of security measures. Organisations, therefore, need more intelligent and adaptive processes for threat detection that can find known and unknown threats.

2.2 Who Is Affected

Cyber risks impact several sectors and actors. Successful cyber attacks lead to operational disruption, data breaches as well as financial losses to the organisations. Government agencies can be breached about critical infrastructure and sensitive national data. Small and medium enterprises of all sizes are subject to risks to IP theft, service interruptions and regulatory compliance failure (Goswami, 2024).

People also lose out when identity theft, financial fraud and privacy theft occur. Society continues to see the impact of cybersecurity failures increase with advancing digitalization.

2.3 Why the Problem Is Important

Failure to detect cyber threats effectively can result in significant financial consequences, including recovery costs, legal penalties and loss of revenue. Ransomware and network intrusions can damage productivity and service delivery within organisations. Security

incidents can also harm an organisation's credit, decreasing customer count and stakeholder confidence (Khalaf et al. 2024).

Taking into account the growing sophistication of cyber-attacks, it becomes vital to improve the system of detecting the threats to make sure that businesses will keep functioning normally, their data will be protected, and they will maintain their resilience.

2.4 Research Gap

Even though machine learning has become a highly topical issue of cybersecurity-related studies, there are certain problems associated with the accuracy, scalability, and adaptability of the systems in question. The existing products are effective in detecting the particular types of attacks or work well in particular circumstances (Katiyar et al. 2024). Moreover, false positive detections are quite common, which makes anomaly detection less efficient.

More studies are required concerning the effectiveness of anomaly detection and machine learning technologies used for threat detection in cybersecurity. The information gained from researching this problem may prove useful in creating better threat detection systems.

2.5 Research Questions

RQ1: How accurately are anomaly detection techniques in identifying cybersecurity threats?

RQ2: What are the best performing machine learning methods for cybersecurity threat detection?

RQ3: What accurate can machine learning be in the detection of new or zero-day attacks?

RQ4: What obstacles and boundaries do pose to machine learning-based anomaly detection systems?

3. Aims and Objectives

3.1 Aim

The aim of this project is to examine how well the techniques of anomaly detection and machine learning can enhance cybersecurity threat detection and assist with malicious activity identification in contemporary digital landscapes.

3.2 Objectives

Objective 1: To analyze and understand the state of cybersecurity threats and traditional threat detection methods.

Objective 2: To explore Anomaly detection principles and uses in a cybersecurity environment.

Objective 3: To analyse machine learning techniques commonly used for cybersecurity threat detection.

Objective 4: To test the Anomaly Detection and Event Classification via Machine learning Techniques to find out the Cyber Threats.

Objective 5: To identify problems, constraints and the future possibilities of an intelligent cybersecurity threat detection system.

3.3 Principal Problem to be Resolved

The principal problem we are tackling with this project is that current cybersecurity detection systems are incapable of effectively dealing with emerging threats and unknown threats. With the sophistication of cyberattacks, organizations are in need of systems that adapt to their unique detection and respond to new attack patterns with more accuracy and efficiency.

3.4 Research Questions to be Addressed

Current research will include four key questions such as efficiency of anomalies detection techniques, possibility to use machine learning technologies in cybersecurity, identification of new threats, and difficulties to use intelligent systems for the purpose of threat detection.

3.5 Approach to Addressing the Questions

This project will be concerned with current research on anomaly detection and/or ML technologies that are being utilized for the sake of Cyber Security. In particular, how the utilization of these approaches in relation to intrusion detection and threat analysis (in this case, especially interesting aspect) can be considered within the framework of: Decision Trees, Random Forest, Support Vector Machines, K-means Clustering, and Isolation Forest algorithms. This research will involve analysis of literature and industrial reports in order to assess the effectiveness of utilization of prevention measures in terms of cyber security threat detection.

3.6 Research Strategy and Methods

The positivist research philosophy will be employed for the objective analysis of the performance of machine learning tools applied to cybersecurity threat detection. The deductive research methodology will be applied for the review of the existing theories/models for anomaly detection and cybersecurity threat detection. Within the context of this study, an experimental research design will be applied for developing machine learning models that will be tested on the basis of the dataset relating to cybersecurity. The data collection process will employ the secondary data collection method, which will consist in the application of the information contained in cybersecurity databases and scholarly papers.

4. Legal, Social, Ethical and Professional Considerations

4.1 Legal Considerations

This research using the application of machine learning and the network traffic data must be carried out in compliance with the legal and regulatory requirements. The data collection and processing process in the study shall use the public cybersecurity data, which have been collected legally and released for the purpose of research. The research shall comply with the relevant data protection laws, such as GDPR and others. The study shall ensure that personal information will not be abused.

4.2 Ethical Considerations

Ethics play a significant role in building machine learning models for cybersecurity. This research will help use data responsibly and keep out activities that may compromise privacy or security. Data sets to be used will be anonymised and from a valid source.

Additionally, the study will take into account the scope for algorithmic bias, which can affect detection outcomes and lead to unequal outcomes. Simplicity of conveying the results of the studies and/or models' performance will be ensured during the project.

4.3 Social Considerations

Cybersecurity impacts persons, organizations, and society. A robust threat detection system can safeguard sensitive data, lessen cybercrime, and build user trust in digital systems. But over-monitoring and algorithmic decision making can cause privacy issues. The project will thus weigh security measures and human rights against machine learning algorithms for detection.

4.4 Professional Considerations

The project will be conducted in a professional manner, following due academic research and cybersecurity practices. Research will be performed with integrity, responsibility and in compliance to institutional ethics. A good referencing process will be followed and results presented will be accurate and respect for intellectual property will be kept during the study.

4.5 Risks and Mitigation

Concerns may be related to the quality of the data, the distribution of the classes, over-fitting of machine learning models, and the inability to recognize new threats. These risks will be addressed by appropriate data set selection, data processing methods and validation methods as well as using multiple criteria for evaluating the models and creating a viable and stable outcome of the project.

5. Background

5.1 Cybersecurity Threat Detection

The term cybersecurity threat detection describes the process of discovery of threats, breach of security or any attempt by an unauthorized person to break into a computer system or network. For most organisations, technological dependency of any kind has become a great concern, and in the scope of technology, cyber security has gained a lot of importance in all the sectors (Inuwa and Das, 2024). Malware ransomware is phishing, distributed denial-of-service (DDoS) attacks, and more are getting worse — and more

common. As with most security software, traditional products such as antivirus and firewalls have their place and offer strong protection, but are less effective at uncovering new and stealthier attacks.

As such, threat detection is now an integral part of today's cybersecurity initiatives. Robust detection systems allow organizations to detect attacks early in the process, mitigate the impact of an attack if one occurs, and enhance their information security response effort. In today's era of abundant network and system-generated data, automated threat detection takes on a major part of importance for security analysts (Sarker et al. 2020). In the light of this, researchers and practitioners have investigated advanced analysis techniques that are able to detect malicious behaviour amongst large and complex data sets.

5.2 Anomaly Detection in Cybersecurity

A very popular way of detecting anomalies is to detect unusual patterns that deviate from normal system behavior. In cybersecurity settings, anomalies can be symptoms of malicious activities or potential internal threats, unauthorized access attempts. Anomaly detection algorithms are based on the idea of identifying deviations from known behavioural patterns; in contrast with signature-based detection methods.

Zero-day attacks and advanced persistent threats have made anomaly detection crucial. They often get around conventional security mechanisms, which is because the signatures will need to be learned by the defenses (Rao et al. 2024). Normal behaviour can be learned to analyze anomalous behaviour by anomaly detection systems, which can detect new threats.

Statistical, Distance-based and Density-based are two main types of anomaly detection techniques, along with machine learning techniques. The statistical analyses detect deviations from known (theoretical) distributions; the clustering and classification detects observations that diverge from normal distributions. Recent studies show that anomaly detection could be of great aid in intrusion detection systems as it helps in the identification of threats earlier and adapts to them more easily.

5.3 Machine Learning Techniques for Threat Detection

Machine learning is proving to be one of the most impactful technologies in cybersecurity today. Machine learning algorithms can analyse large volumes of data, identify hidden patterns and automatically detect suspicious activities with minimal human intervention. Machine learning is an ideal fit for today's complex cybersecurity environment and is valuable in situations where data volumes become harder to handle with manual analysis (Jain and Mitra, 2025).

In Cybersecurity, datasets are often available together with examples of attack and normal traffic, and are typically dealt with using supervised learning techniques. Algorithms such as Decision Trees, Random Forests, Support Vector Machines and Logistic Regression have been widely used for intrusion detection and malware classification. These methods can be used to gain insights into how features in the input flow are related to categories of attacks so that future observations can be classified accurately.

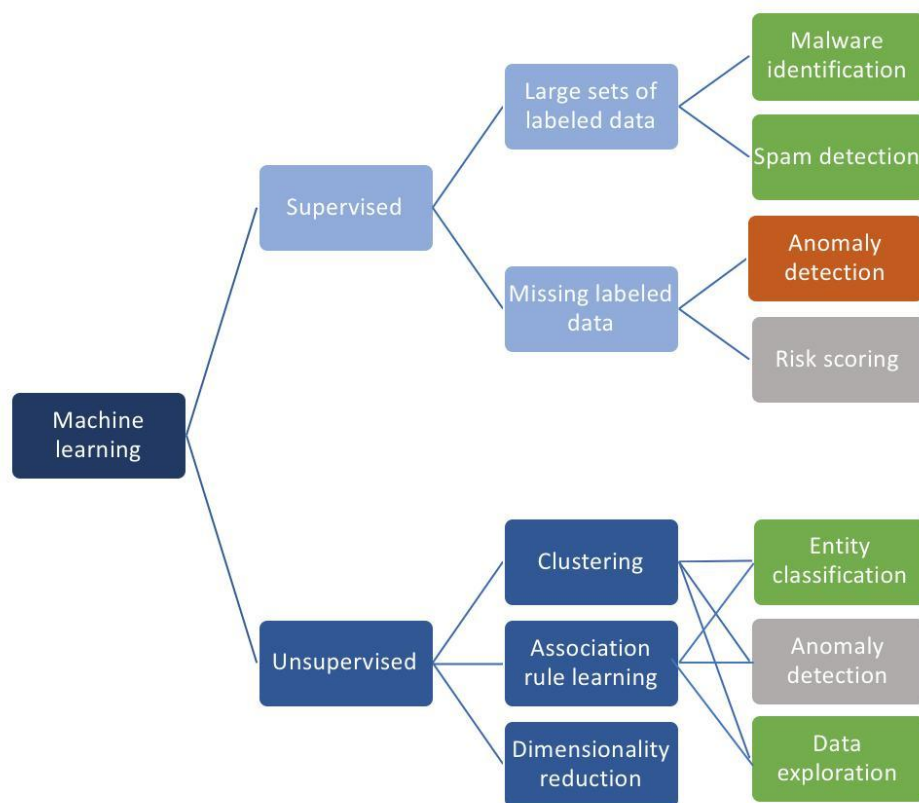


Figure 3: “Machine Learning Algorithms for Threat Detection”

(Source: Jain and Mitra, 2025)

With no labelled data available, unsupervised learning methods have proven valuable concepts. Anomalies can also be detected using techniques like K-Means Clustering and Isolation Forest which can identify observations that are far from the normal pattern. These techniques are particularly useful for the identification of unknown attacks and new threats.

Some of these techniques used in deep learning recently are: Artificial neural networks, Recurrent neural networks and Autoencoders. These methods may be applied to data in cybersecurity information that is complex, and to identify patterns of behaviour that prove to be more effective than traditional behaviour.

5.4 Current State of the Art

Advanced machine learning and AI techniques are increasingly being used with anomaly detection to form the latest in cybersecurity threat detection. Modern security systems employ automated analytics which can handle enormous amounts of network data in real-time. Today, SIEM platforms incorporate machine learning models as a means of aiding proactive threat identification and incident response (Hariharan et al. 2020).

Hybrid methods that fuse several algorithms to optimize detection accuracy and minimize false alarms have been developed. Cybersecurity datasets have shown good performance in ensemble learning methods like gradient boosting (GB) and random forest (RF) (Ofoegbu et al. 2024). Likewise, deep learning architectures have exhibited great promise for identifying potential patterns of attacks that may be hard to detect with traditional methods.

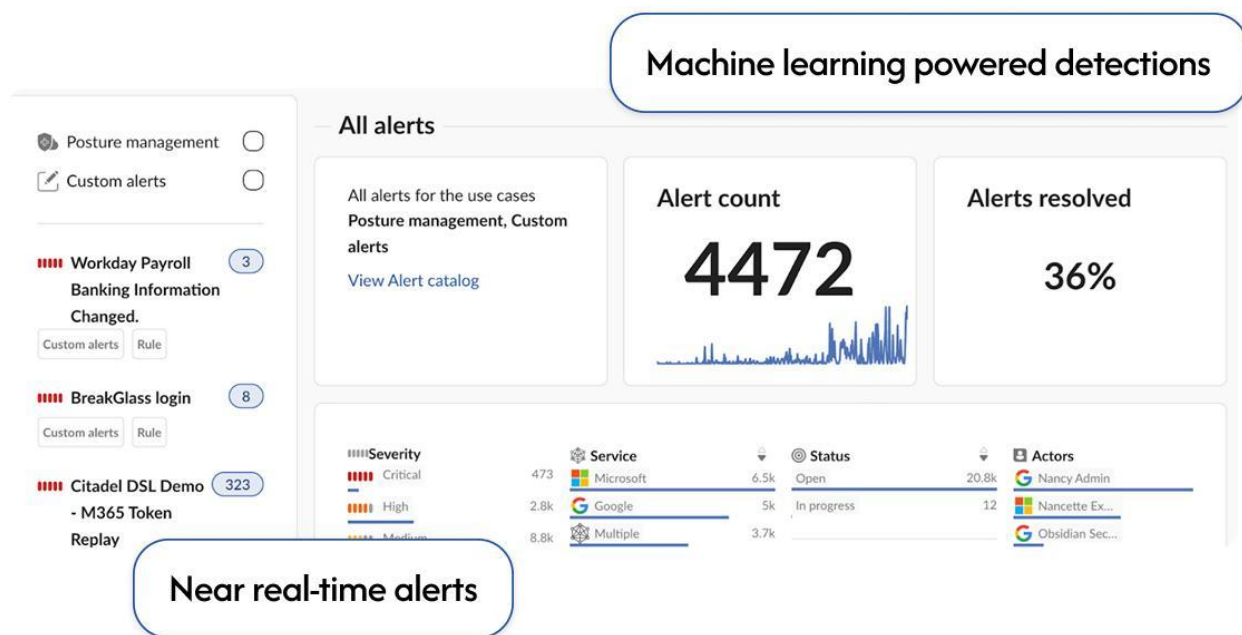


Figure 4: “Modern Security Operations Center (SOC) utilizing machine learning and real-time analytics for proactive threat detection and response”

(Source: Ofoegbu *et al.* 2024)

Perhaps one of the more noteworthy innovations is the incorporation of explainable AI (XAI). Explainability helps security personnel understand what drove a model to label an activity as bad, promotes greater trust and better decision making. These are indicative of the ongoing evolution of smart Cyber Security solutions and showcase the importance of machine learning in the Cyber Security sphere (Saabith et al. 2023).

5.5 Existing Research and Literature

In the last decade, machine learning for cybersecurity has received significant research interest. There have been a number of studies that assess the effectiveness of supervised and unsupervised learning for IDS and threat classification. Previously it has been shown that ML techniques could be more accurate in determining the attack, especially in case of an anomaly, compared to the traditional signature-based technique (Bin Sarhan and Altwaijry, 2022).

There are various open-source databases utilized for experiments with the ML-based techniques including NSL-KDD, CICIDS2017 and UNSW-NB15. There are several relevant papers devoted to the use of different algorithms including Isolation Forest, SVM

and Random Forest, in these datasets. In addition, there is a research dedicated to deep learning approaches to classify other issues caused by complex attacks (Alsalman, 2024).

However, there are some disadvantages that should be pointed out; for example, too many false positives, complicated modeling, class imbalance and computer requirements are still the main problems. Furthermore, a model trained on a particular dataset cannot be generalized in practice.

5.6 Relationship Between the Proposed Project and Existing Work

The project under discussion has been developed as a continuation of the research on anomaly detection and the applications of machine learning in Cybersecurity conducted in the past. According to the findings made by researchers, there is a positive impact of machine learning algorithms on threat detection accuracy; however, the issues related to scalability, generalization, and performance require further solving (Shaukat et al. 2020).

In terms of the novelty of the study, the project is a further development of the past research since it implies the use of cybersecurity dataset for comparing machine learning techniques. The study is going to examine the role played by anomaly detection methods in threat detection. As mentioned above, the concept is not novel but rather a continuation of the past research on the applications of machine learning in practice (Ameen, 2021).

The study will build upon existing knowledge in examining algorithm performance, strengths and weaknesses, and whether machine learning approaches are suited to modern-day environments for threat detection.

5.7 Industry Relevance and Potential Contribution

Security in cyberspace is a high-profile issue for governments, businesses and critical systems operators around the world. More and more, organisations need intelligent security solutions that can detect advanced attacks, and at the same time process a huge amount of data flowing through the network (Ahsan et al. 2022). For this reason, the need for machine learning empowered threat detection systems is still rising in different industry sectors.

The outcomes of this project could have practical implications for future research on the success of anomaly detection and machine learning approaches for cybersecurity

applications. The study could help organisations choose appropriate algorithms to use in the threat detection process by comparing performance across varying algorithms (İsa, 2023). Moreover, the project helps continue the academic and industrial debates on the future of AI in cybersecurity. The outcomes can be utilized in the future development of automated systems for monitoring for security, adaptive and intelligent intrusion detection, and cyber defence systems. Reference

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Appendix B: Artefact/Dataset

<https://www.kaggle.com/datasets/jancsg/cybersecurity-suspicious-web-threat-interactions>